



UNIVERSITY OF THE EAST
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Methods of Research

**An Internet of Things-Based Framework for Cardiac Arrhythmia
Detection via Photoplethysmography and Machine Learning**

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

Cardiovascular diseases are diseases caused by some sort of anomaly in the regulation of blood pressure, blood vessels of the heart, blood vessels of the brain, and in the muscles of the heart. Hypertension, Ischaemic heart disease, stroke and heart failure are examples of such diseases. (Centers for Disease Control and Prevention 2023; Benjamin et al. 2019). Despite significant advancements in healthcare technology, cardiovascular diseases remain a leading cause of morbidity and mortality worldwide. Notable examples of such advancements include Electronic Health Records (EHR) and Artificial Intelligence (AI), which have substantially transformed the ways in which diseases are analyzed and managed. These diseases, behaviors, and demographics are all taken into account when collecting data. Such data is vital when trying to understand the trends of diseases, how to best help a patient in a clinical setting, and in disease prevention, and/or managing strategies. (World Health Organization 2023; Global Burden of Disease Collaborative Network 2020). An example of the trend we speak of is that in 2023, approximately 19.2 million people died as a result of cardiovascular disease. This statistic is nearly equal to a two-fold increase in susceptible cases starting from 1990. This increase is primarily attributed to a rapidly growing and aging population. The worldwide burden is driven chiefly by a lack of attention to avoidable risk factors that make up a person's overall cardiovascular health (CVD) - e.g., hypertension, unhealthy diets, high levels of LDL cholesterol and pollution in the environment. CVD contributes significantly to high levels of Disability Adjusted Life Years - DALYs. Worldwide, CVD continues to be the leading cause of death for people and there are millions

of deaths from CVD each year. Although both the age-standardized mortality rates and the prevalence rates of CVD appear to be steady, others believe that the lack of changes in these measures indicate that current methods of preventing CVD (e.g., screening and modifying risk factors) have had little impact (Roth et al., 2020; WHO, 2023). Most of the new deaths from CVD have occurred and will continue to occur due largely to changing national demographics and an increasingly aging population globally (United Nations, 2022). As a result of age and the associated biological changes, an older person often has several risk factors for developing cardiometabolic disease (CMD). Given this trend toward aging, there is a growing demand for long-term management of CMDs and risks, with a special emphasis on secondary prevention and treatment of atherosclerotic diseases (e.g., coronary artery disease and stroke) (Virani et al., 2021).

Arrhythmias are an increasingly important segment of the global burden of cardiovascular disease, (Lippi et al., 2021; Kingma et al., 2023) and create a significant impact on health care because they may lead to stroke or disability and ultimately through syncope or sudden cardiac arrest (tachyarrhythmia or ventricular fibrillation). In 2019, there were 4.7 million new AF cases, with 117,000 deaths directly attributed to AF. The effect of AF is very much felt in terms of disability-adjusted life years (Kingma et al., 2023; Xu et al., 2022). AF is more prevalent in the elderly and among males as well as developed countries in terms of socioeconomic status. AF is significantly associated with many prevalent and treatable conditions, including hypertension, obesity, and diabetes as well as behaviors such as alcohol use and tobacco use (Kingma et al., 2023). In addition to thromboembolic complications from AF, AF contributes to the progression of heart failure, increases hospital

admissions, and increases rates of sudden cardiac death (Lippi et al., 2021; Gopinathannair et al., 2020). As documented by McJunkin and Wilken (2023), clinical outcomes and management strategies vary considerably across geographic regions. High-income countries demonstrate superior outcomes owing to their access to anticoagulation medications, catheter ablation procedures, and implantable devices; conversely, many low-income countries exhibit markedly poorer management, characterized by limited specialist capabilities and restricted access to implantable devices or even basic therapeutic options. The absence of equitable access to appropriate care has created substantial gaps in the needs of those with significant dysrhythmias, underscoring the necessity of developing cost-effective, inclusive interventions aimed at understanding the root causes of these disparities and identifying the various management patterns within each country (Lippi et al., 2021; Kingma et al., 2023). While there is some historical and hospital-based data, there is limited data on the local clinical value of the local cardiac health ecosystem. Atrial Fibrillation (AF) is the most common sustained cardiac arrhythmia in the world (Lippi et al., 2021; Kingma et al., 2023). However, there are some local data on the prevalence of this condition. Previous national data, ECG, record data that the prevalence of AF in adult Filipinos is 0.2 percent. However, older age groups show a significantly higher rate (Fadreguilan, 2023). This low prevalence is, however, expected to increase AF and other rhythm disorders. This will be the case because of the aging population in the country and the increase in the prevalence of comorbid cardiovascular risk factors such as hypertension and diabetes. AF and other arrhythmia increase the risk of stroke and systemic embolism which is a significant public health concern especially in place with limited screening and treatment options (Fadreguilan, 2023; Asia Pacific Heart Rhythm Society 2022). Hospital admission data from PhilHealth shows

that the number of hospital admissions for cardiac arrhythmia which required the implantation of a pacemaker shows both demand from the health system and a limited number of higher-level health services. Bradyarrhythmias, complete A-V Blocks and Sick Sinus Syndrome (SSS) typically occur in older adults who have permanent pacemakers (PPMs) implanted due to abnormal electrical conduction in the heart. There are many more cases of these arrhythmias that are detected/followed-in urban settings than in rural areas, reflecting the lack of access to specialty cardiac care in rural areas (Philippine Heart Association 2025). The existence of many documented cases of uncontrolled bi- or tri-affected electrical conduction problems indicates an urgent need to develop better methods for the long-term monitoring and early detection of all types of arrhythmias. Some of the primary issues that many primary care providers report are: Low level of awareness surrounding arrhythmia; absence of routine electrocardiograms (ECGs); lack of cardiology availability; and lack of knowledge on detection/management, all result in difficulties in the early diagnosis and effective treatment of patients with arrhythmia. Collectively, these issues can lead to an under-appreciation of the high burden of arrhythmias on society and may lead to many issues that are easily preventable, such as sudden cardiac death. There are no known programs in rural communities utilizing telemetry and/or remote monitoring devices or outreach services that could improve the early detection of arrhythmias. To enhance an individual's ability to recognise an arrhythmia in the community, it is imperative to strengthen the Community First-Level Care (CFLC) system and provide diagnostic ECG capability to CFLC providers. The best way to improve the understanding of arrhythmias in both patients and their providers is to increase the education of patients and providers regarding arrhythmia. Collectively, these deficiencies highlight the critical need to develop

comprehensive, integrated strategies for the monitoring and management of arrhythmias, with the goal of reducing cardiovascular morbidity and mortality in the Philippines.

One of the most challenging aspects of managing cardiovascular disease in the Philippines is the number of 'silent' cardiac diseases that people have. Many people have cardiac diseases that do not have any visible symptoms for long periods of time, which leads to the misdiagnosis of many patients. Due to the lack of overt symptoms displayed by patients and lack of easy access to appropriate screening tools, it is reasonable to conclude that there is a significant underestimate of the burden of cardiovascular morbidity. In addition to the lack of overtly symptomatic individuals in the Philippines, there is also a complete deficiency of the infrastructure necessary to identify individuals who are at greater risk for developing clinical cardiovascular disease. Although there are medical professionals capable of providing the necessary screening and diagnostic tests and medical devices capable of accurately identifying patients with hypertension, the availability of appropriate medical devices to provide necessary determinations of cardiovascular disease is not available. As a result, individuals with these clinical diseases will not be diagnosed until they are much further along in their progression and have already suffered from significant sequelae including stroke. The absence of clinical intervention leading to the death or disability of individuals with cardiovascular disease is one of the most significant factors that creates problems with the epidemiology.

The epidemic of cardiovascular disease is considerably underestimated because of the inability to provide adequate surveillance for the disease. Due to an absence of access to proper medical treatment, large quantities of people in the Philippines die either at home or

without proper medical assistance and may thus be deemed not to have died from an arrhythmia or an underlying cardiovascular condition. There is insufficient use of electronic health record (EHR) systems across hospitals. Many still utilize paper records, creating an overall problem to accurately document death. Due to the lack of familiarity or knowledge with the local population, other associated comorbidities (e.g., Sleep Apnea) related to arrhythmias are also often undiagnosed, untreated, and under-researched (Chesky et al. 2022). The total underreporting of fatal cardiovascular events and associated underlying causes of cardiovascular events displays the level of need for a widely used monitoring solution to collect real-world health data. Although there has been an incredible amount of new technology created for use in wearable devices, massive amounts of funding have been absent for developing and hard-to-reach places with extreme data gaps. Systematic reviews demonstrate that newer health monitoring devices such as biosensing patches and smartwatches have implemented excellent remote patient monitoring and arrhythmia detection technologies to help in the management of these types of medical conditions. In general, commercially available devices cost too much money, as well as require continuous access to an Internet connection to function. For this reason, it can be said that the majority of people in the Philippines are unable to purchase them (Sultana et al., 2025). Hence, developing low-cost, autonomous monitoring systems is prone to bridge the effort to bridge the gap between technologically advanced diagnostic systems and local health systems. Research into PPG (photo plethysmography) is still lacking according to Neha, et al, (2023), who also say that current research methods are "black box models" due to their systematic processes used to interpret the results. Thus, they create barriers for capturing data during arrhythmia episodes. The need for clarity and clinically relevant techniques for detecting

arrhythmias using PPG signals is an area of urgent concern, particularly when it comes to the rationale of methodologies used for detecting atrial fibrillation. The study has identified an approach to solve this need by utilizing deep machine learning techniques with the PPG-Beats toolbox as a means of detecting atrial fibrillation and will provide a newly designed clinically actionable and accurate method of using PPG signals to detect arrhythmia. Although there is a significant increase in the number of IoT-based heart monitoring systems being developed, it is still difficult to establish a reliable means of using them for screening individuals for their risk of developing arrhythmia. Research conducted by Kumar et al. (2021) showed that IoT-based cardiac monitoring systems that utilize machine learning were capable of accurately classifying arrhythmia that was recorded through the system. However, most systems generally have cloud-based architecture for processing the data, which could delay the results that would be available for clinical patient care. Furthermore, as shown by Sharma et al. (2022), many Internet-based IoT systems for arrhythmia detection require continuous access to the Internet for their optimal performance. This limits their use in developing countries. Most current techniques for detecting arrhythmias are designed to identify arrhythmias after an abnormal heart rhythm has been detected. A restricted ability to monitor risks throughout the IoT (low risk) could suggest that predictive modelling methods could be developed with greater complexity using a predictive model that will maximize the potential of IoT networks. In addition to their ability to perform predictive analysis of data, IoT healthcare systems face challenges associated with the security and privacy of patients' data. There is a tendency amongst researchers in the field of IoT healthcare to focus on improving system performance without offering proposals for implementing data integrity solutions: machine learning based IoT healthcare systems have

shown that performance enhancements take priority over implementation of data security methodologies. Medical sensors and the data they provide may be vulnerable to unauthorized access and/or tampered data; this is a particular concern regarding data produced by continuous monitoring (e.g. electrocardiograms (ECGs)). Numerous researchers have proposed an approach whereby medical sensors are integrated with blockchain technologies to ensure that data captured by medical devices is secure and free from tampering. The first working example of an IoT healthcare system based on blockchain technology was developed and validated with initial evidence of enhanced performance and traceability for the data associated with this system. This first example, as well as subsequent developments using similar designs, will ultimately allow for greater data integrity and traceability within systems utilizing IoT technology to provide healthcare services. However, many systems designed for the healthcare sector using blockchain associated technologies have not yet been fully developed or are at the theoretical stage only. Few studies have demonstrated the use of blockchain technology for real-time monitoring systems that can be used to detect arrhythmia; additionally, the complexity of implementing blockchain technologies can create an extraordinary demand for computational resources on edge devices (Nguyen et al. 2021). This is often a challenge to the security of the blockchain as well as providing an efficient response time to the user; there are similar ongoing challenges with integrating blockchain-based solutions with other IoT devices and edge nodes. Clarity regarding patient consent and access control mechanisms also presents obstacles as they relate to establishing trust in a clinical setting and the willingness to utilize these technologies. More research is warranted to better understand the interrelatedness of security, effectiveness, and usability within IoT-based solutions within healthcare environments.

The aim of this study is to create affordable, easily accessible and reliable methods to identify arrhythmias and other heart diseases (i.e. cardiovascular disease) as early as possible before they develop into grave or fatal risks. When an arrhythmia (such as atrial fibrillation) is detected early enough to allow for medical intervention on that arrhythmia, then the patient has almost a 50% higher likelihood of being able to prevent a stroke, heart failure or sudden cardiac arrest. Although there are many portable/mobile technologies currently available, most of them are either far too expensive to readily access or employ too complex a technological approach to readily utilize (Nguyen et al. 2021); this is particularly true in poorer regions throughout the Philippines and all of Asia. Therefore, the plan of this research study will be to adapt existing low-cost (cost-effective) body-worn devices and mobile ECG technologies to provide high-quality, always-on, accurate and real-time monitoring of heart rhythms (heart beats) to allow use in low resource environments. Additionally, the creation of user-friendly interfaces to be used with these new physiological monitoring devices will allow caregivers other than traditional medical providers (such as family and friends) and patients to have access to advanced diagnostic equipment in their own communities; this will provide continuous remote monitoring outside of hospital settings, accessible to both at-risk and general adult populations. This study ultimately aims to decrease the frequency of underdiagnosis (e.g., missing undiagnosed/asymptomatic), or incorrectly diagnosed arrhythmias that are either transient or present. The emphasis of this study on educational programs for both patients and healthcare providers and the need for proactively assessing patients for the development of arrhythmias. This study is also part of the research project connected to the 3rd Sustainable Development Goal (SDG), which deals with ensuring healthy lifestyles and providing health benefits for everyone at all ages, specifically target

3.4, which is related to decreasing mortality rates due to non-communicable diseases (NCD) by increasing focus on prevention and early treatment (United Nations 2022). Furthermore, this study intends to make arrhythmia detection technology more accessible to a greater number of patients. Thus, this research addresses the gap in the equity of health services between urban and rural populations and aims to contribute to Universal Health Coverage (UHC) Philippines (Philippine Heart Association 2025). The intention of this study is also to increase the integration of data from wearable technology into the healthcare system in order for practitioners to monitor their patients and develop effective interventions to control the level of demand by the healthcare system. This framework will provide nearly real-time information about the expected patient population with arrhythmia to help healthcare decision-makers and health planners to solve the issue of arrhythmia care delivery. Ultimately, this study will provide a better result for patients.

Statement of the problem

Despite the growing body of literature on wearable-based cardiac monitoring and machine learning-driven arrhythmia detection, no existing study has developed and validated a unified framework that simultaneously integrates PPG-based Atrial Fibrillation (AF) detection, edge-level deep learning classification, and blockchain-secured data management within a low-cost, offline-capable IoT system designed for the Philippine primary care setting. While studies such as Kumar et al. (2021) and Pedrosa-Rodriguez et al. (2024) have demonstrated the feasibility of IoT-enabled PPG systems for AF detection, these systems largely depend on continuous cloud connectivity, rendering them impractical for rural and

resource-constrained communities in the Philippines where internet access remains inconsistent and specialist cardiac care is scarce (Sultana et al., 2025; Philippine Heart Association, 2025).

Furthermore, although blockchain technology has been proposed as a mechanism for securing health IoT data (Egala et al., 2021; Sumathi et al., 2022), its integration into real-time, edge-deployed arrhythmia detection systems remains largely theoretical, with few working prototypes validated under real-world clinical or community conditions (Nguyen et al., 2021). Locally, the burden of AF among Filipinos is compounded by the absence of routine electrocardiogram screening, limited cardiology availability in rural areas, and a significant reliance on paper-based health records that impede accurate disease surveillance (Fadreguilan, 2023; Philippine Heart Association, 2025).

The consequence of these overlapping gaps — technological, infrastructural, and clinical — is a persistent and dangerous underdiagnosis of AF in a population at growing risk due to aging demographics and rising cardiovascular comorbidities such as hypertension and diabetes (Kingma et al., 2023; Roth et al., 2020). This study therefore aims to address this gap by proposing, implementing, and evaluating an IoT-based framework for cardiac arrhythmia detection via photoplethysmography and machine learning — a system designed to be affordable, edge-capable, secure, and contextually appropriate for deployment within the Philippine community health system in support of SDG 3 and Universal Health Coverage (United Nations, 2022).

The purpose of this research is to form a framework using the Internet of Things (IoT) that can aid in the early identification of cardiac arrhythmia, particularly atrial fibrillation (AF). In addition, this study is intended to provide reliable health information to

the Filipino community, where there is currently a lack of diagnosis for many individuals at risk due to the absence of symptoms, lack of available screening and healthcare services, and prohibitive costs associated with these services.

The proposed solution is intended to close the gap between the current highly advanced, expensive medical devices available and those available to the majority of Filipinos by combining low-cost wearable devices and an IoT-based, clinically actionable, and machine-learning (ML) framework founded on blockchain technology. These elements are all aimed at providing support for the attainment of SDG #3 by contributing to a decrease in preventable mortality in developing countries.

Overall, this research is focused on the creation and evaluation of several different solutions to address the issue described above. Researchers will be guided by the following questions to achieve this objective:

Specific Problems

To answer this question, this research answer will answer the following questions:

1. What photoplethysmography signal inputs and physiological parameters are required to enable reliable detection of atrial fibrillation using a photoplethysmography-beat detection framework?
2. How can an Internet of Things-enabled device be designed and implemented to acquire photoplethysmography signals and process them through machine learning techniques for atrial fibrillation detection?

3. How can an interpretable machine learning approach be integrated into the system to classify atrial fibrillation based on photoplethysmography-beat features while ensuring real-time and edge-level processing?
 - Specifically, how can Gradient-weighted Class Activation Mapping (Grad-CAM) be utilized to highlight the PPG signal segments most influential to the model's AF classification output?
4. What system outputs and functionalities can be produced to support early atrial fibrillation detection, user notification, and clinical decision support in non-hospital settings?
5. What is the performance and overall quality of the developed system when evaluated using the ISO 25010 software quality model?

How does the system perform in terms of the following sub-characteristics:

- 5.1 Functional Suitability: Does the system accurately identify AF and provide the intended diagnostic support?
- 5.2 Performance Efficiency: Does the edge-level processing maintain low latency during signal classification?
- 5.3 Reliability: Can the device provide continuous heart rhythm detection in low-resource settings?
- 5.4 Security: Does blockchain integration effectively manage access control and patient data privacy?
- 5.5 Usability: Is the interface sufficiently user-friendly for non-specialist healthcare providers and patients?

- 5.6 Maintainability and Portability: Is the system sustainable for long-term use within primary care workflows?

Significance of the study

The research provides significant value to a variety of organizations and also to many members of the community. The multiple benefits of early detection of atrial fibrillation and community-level cardiovascular care are apparent at a variety of levels, including practical, clinical, technological and at the policy level.

Department of Health of the Philippines.

This study aims to help the Health Department develop an economical, Internet of Things (IoT)-based atrial fibrillation (AF) detection program that can enhance current national cardiovascular disease (CVD) surveillance and early detection programs. The new technology will also help identify individuals at high risk for developing CVDs, particularly those from under-represented populations; support prevention strategies; and reduce the long-term costs associated with stroke and heart failure for publicly funded health care systems.

Local Government Units and Primary Health Care Centers.

Local Governments and Healthcare facilities, such as Barangay Health Centres, will be able to conduct outreach screening for Atrial Fibrillation (AF) easily and affordably thereby decreasing their reliance on hospital-based diagnostic tools and providing a means of reducing their workload and providing better referrals for patients identified as needing additional evaluation by a physician.

Healthcare Professionals and Primary Care Providers.

There are many possibilities for healthcare systems to make use of continuous heart rhythm data, obtained out of hospital, and through the use of machine learning, to allow for accurate interpretation. One of the most beneficial attributes of this type of heart rhythm interpretation is that it allows healthcare practitioners' use of the data generated to make clinical determinations i.e. to make an informed decision, as it minimizes the amount of reliance they must place on complex, unusable models. As a result, healthcare practitioners will have greater confidence in their diagnosis and increase the likelihood they will be able to act quickly on findings from the heart rhythm data collected/obtained.

Patients and At-Risk Individuals.

Atrial fibrillation (AF) can be recognized and treated early without requiring frequent visits to the emergency room by providing ongoing monitoring with instant notifications and an increased ability for the patient to take responsibility for their own cardiovascular health. Patients now have less risk for sustaining serious complications from undiagnosed or intermittent arrhythmias due to improved methods of identification of AF compared to previous years.

Researchers and Academic Institutions.

This project adds to what has already been published about how to find heartbeats using photoplethysmography, machine learning and IoT. In addition to creating clinically

actionable detection systems for detecting arrhythmias, any future work with community-engaged digital health technologies could use this work as a foundation.

Future Developers and System Designers.

The base framework designed for system developers and designers provides a way to measure and/or integrate other types of cardiovascular illnesses and/or integrate new technologies into their existing application. The design of the modular system along with an evaluation framework based on ISO/IEC 25010 allows for a systematic approach to build scalable, standards compliant (ISO 25000 Series) health care monitoring systems.

Scope of the Study

- This study focuses on the design, development, and evaluation of an Internet of Things-enabled prototype system for atrial fibrillation detection based on photoplethysmography signals and machine learning techniques.
- The acquisition of photoplethysmography signals using affordable wearable sensors, as well as signal preprocessing and heartbeat detection to generate beat-to-beat intervals and heart rate features.
- The development of an Internet of Things architecture that enables real-time data acquisition, local processing, and transmission of photoplethysmography data from the wearable device to a monitoring interface
- Implementation of an interpretable supervised machine learning model for atrial fibrillation classification based on features extracted from photoplethysmography-beat data.

- The development of system outputs such as real-time alerts, visualizations of heart rhythm trends, and decision-support indicators intended to guide users in seeking timely medical consultation.

Limitation of the Study

- This study is limited to the detection of atrial fibrillation and does not include the classification of other cardiac arrhythmias such as atrial flutter, ventricular arrhythmias, or bradyarrhythmia.
- Photoplethysmography signals may be affected by motion artifacts, skin tone variability, and sensor placement, potentially impacting detection accuracy under certain conditions.
- This study does not replace clinical-grade diagnostic tools such as electrocardiography. The proposed system is intended solely as a screening and decision-support tool and not as a definitive diagnostic instrument.
- The findings of this study are derived from a pilot deployment conducted within Caloocan City and may not be directly generalizable to rural provincial settings in the Philippines, where infrastructure capacity, population demographics, and health literacy levels differ significantly from those of urban communities.

CHAPTER 2

REVIEW OF RELATED LITERATURE

1. Identification of Physiological Parameters of Interest and PPG Signal Inputs for AF Detection

Evidence from research presented in the paragraphs above suggests that PPG signals and measures can be useful as inputs to detect atrial fibrillation (AF) in a manner that is dependent on beat detection by accurate processing of signals, beat interval extraction from PPG signals (Millán et al., 2020). In a systematic review of how extracted features from PPG signals can be applied to AF detection, Millán (2020) identifies that most published studies on AF detection validate the presence of AF through statistical rhythm descriptors calculated

from PPG beat-level signals (beat-to-beat interval, inter-beat interval, or from interval irregularity), or through quantification of the variability between beats or pulse rate (Pereira et al., 2020). In a comparison of the processing results of PPG data to ECG (electrocardiogram) reference data from the studies included in the comprehensive systematic review by Millán, the authors report that the PPG-derived pulse intervals and waveform characteristics were informative indicators of AF as compared to ECG, thus beat timing and beat variability as measured by PPGs can serve as valid physiological markers of AF. Furthermore, a recent study that evaluated PPG for the assessment of AF in a large, real-world cohort using smartphones demonstrated that PPG can be used to assess rate and rhythm of AF during normal outside of clinical settings (Gruwez et al., 2024).

The use of wearable and wrist-based monitors to determine intervals and interval variances in relation to how the device detects AF has been reported by different authors. One author, Bashar et al. (2019) demonstrated through their research regarding the detection of AF using wrist-based PPG from smartwatches, that wrist-based PPG analysis of beats can also be used for automated AF classification. Another group of authors, Liao et al., indicated that the duration of the sample periods as well as the interval variability of the recorded beats are key to detecting AF through wrist PPG and that the PPG must have stable AF detection and a segment length that is sufficiently long to detect AF. The use of a PPG-based arrhythmia classification system in conjunction with a timing feature(s) and a set of other physiological parameters will increase the level of accuracy associated with AF classification, based on a database of PPG arrhythmia classifications.

Morphology of the waveform and indices of signal quality are included as part of the framework to detect atrial fibrillation. They have been shown to be significant components of atrial fibrillation detection frameworks. Specifically, Yan et al. demonstrated how using a method of assessing the quality of the waveform of the signal allows for the removal of unreliably represented segments of data when extracting PPG signals to improve the accuracy of AF detection. Ding et al. used strong learning methods, developed from PPG inputs, to detect AF and used validated extraction methods of beats and noise-aware pre-processing approaches as the most effective methods for determining the PPG input-based AF detection. Other studies have indicated that accurate identification of AF from PPG signals is not achievable unless all relevant input parameters, including: beat-to-beat intervals; metrics of interval variability; descriptions of morphological characteristics of the waveform; signal quality indices, are accurately defined.

2. Atrial Fibrillation Detection by Using Machine Learning Techniques and Photoplethysmography Systems Enabled by IoTs

The development of an Internet of Things (IoT) connected device to collect PPG signals using machine learning (ML) techniques for the detection and diagnosis of atrial fibrillation (AF) is structured in multiple parts, which allow for the integration of the benefits from both the IoT system and the current methods of analysis. First, an IoT device was developed to facilitate real-time collection of PPG signals as shown by research performed by (Pedrosa-Rodriguez et al., 2024). An IoT device will allow for continuous monitoring and real-time collection of PPG, thereby making them very effective in detecting and monitoring AF. In addition, separate research performed by (Aliamiri & Shen, 2024) shows there are

many commercially available IoT wrist-worn PPG collection devices (e.g. smartwatches, fitness trackers) that have integrated PPG sensors and sufficient computing capabilities to give consumers an easily portable, non-invasive, and low-cost means of monitoring for AF (Cinotti et al., 2024) (Pedrosa-Rodriguez et al., 2024). Research conducted on PPG by (Pedrosa-Rodriguez et al., 2024) indicates that non-invasive data collected from PPG has a direct correlation to AF occurrence and diagnosis.

3. Interpretable and Edge-Optimized Machine Learning Classification of AF from PPG Data

Utilizing machine learning techniques, the new framework will develop highly interpretable models as well as optimize edge performance through edge-based data analysis algorithms.

Currently, there are potential research studies proving the feasibility of using machine learning models combined with PPG data for AF detection (Zhang et al., 2023) (Salvi et al., 2024). Both studies demonstrated that thus far, beat-level PPG data can be classified using various supervised machine learning algorithms, including support vector machines, random trees, and lightweight neural networks. Additionally, they have shown the ability of classifier methods to convey AF characteristics such as the beat interval configuration (timing of sequential beats), morphological pulse characteristics, and heart rate variability (HRV) features associated with the PPG signals received. Lastly, results with the PPG model will show predicted clinical implications of how the aforementioned quantitative and qualitative

beat characteristics of a subject enable a classifier to learn and make clinical predictions associated with previously undiagnosed AF. The beat characteristics of the PPG signal provide a clearly defined interpretative feature for machine learning classifiers and form the basis of the proposed framework's architectural and functional design.

Santala et al. (2022) and Zhou et al. (2022) demonstrate that neural networks and attention-based compressed models are able to classify atrial fibrillation while meeting the operational constraints of wearable and edge device technology. The machine-learning models built on optical (PPG) data maintain similar performance when completing real-time operations. (Zheng et al. 2023) and (Sideswar et al. 2021) validated the possibility of using PPG as inputs to support machine learning solutions for real-time applications, and they propose lightweight, efficient inference pipelines for ultra-edge devices. The results show that there is a practical solution for the automatic monitoring of cardiac rhythms through the use of efficient machine-learning solutions and through the use of wearable sensors that provide continuous tracking. Additionally, studies (A. Bench et al. 2024) and (Howlader et al. 2025) confirm that machine-learning-based solutions created with smartwatch data (PPG) can be successfully applied in real-world settings. Employing ensemble learning, supervised classification, and uncertainty estimation methods, the authors found PPG signals from wearable devices to be suitable.

3.1 Using Grad-CAM to visualize key PPG segments driving atrial fibrillation classification decisions

A significant hurdle in using deep learning models (like 1D-CNNs) for wearable health monitoring devices is that they tend to be "black-box" models, meaning that there is usually

little to no indication of how they create their clinical predictions, which limits their ability to be used for widespread clinical practice. To begin bridging this gap and to generate confidence in the automated diagnoses made by these models, Explainable AI (or XAI) techniques such as Gradient-weighted Class Activation Mapping (or Grad-CAM) can be used to provide an adequate amount of interpretability of the model's predictions (Adadi & Berrada, 2018). Grad-CAM was first used for analyzing 2D images, but several experimental studies have shown that Grad-CAM can be used very effectively for interpreting 1-D time-series of physiological signals such as ECG and PPG waveforms (Jahmunah et al., 2022).

The Gradient-weighted Class Activation Mapping (Grad-CAM) algorithm utilizes the last layer's convolutional features from deep neural networks or any machine learning (ML) methods as the foundation for understanding how that ML method arrived at its classification decision.

Grad-CAM provides an empirically derived visualization of the important regions of each time step within the time-series data set (raw PPG or ECG signals) that were considered critical for developing the model's classification. Liu et al. (2022) clearly show how Grad-CAM visualizations highlight the regions, or patterns, or timeframes that produced an atrial fibrillation (AF) classification. Furthermore, there is substantial evidence in the literature that indicates the segments highlighted through the use of Grad-CAM correspond significantly to the morphological characteristics of AF that are used as a basis for making clinical diagnoses through current diagnostic clinical standards. For example, retained

artifacts have provided strong evidence to confirm that the model has localized the key morphological characteristics of the P-wave with respect to the absence of a distinct P-wave, T-wave abnormalities, and irregularity in the inter-beat (R-R) intervals (Luo et al., 2025).

Ultimately, by providing these visual explanations, Grad-CAM translates opaque algorithmic computations into intuitive, clinically meaningful data. This interpretability validates that the neural network is relying on appropriate physiological markers rather than motion artifacts or noise. Furthermore, utilizing interpretable machine learning approaches on PPG data significantly supports the deployment of reliable real-time applications and lightweight inference pipelines on resource-constrained ultra-edge devices (Zheng et al., 2023; Sideswar et al., 2021).

4. Outputs of the System and Functionality of Decision Support for Early Detection of Atrial Fibrillation

Mobile health (mHealth) technologies, wearable devices, and remote clinical decision-making have shown great promise for supporting the early detection of atrial fibrillation (AF) and providing timely alerts to users outside of hospitals. According to Baman et al. (2022), photoplethysmography (PPG) and electrocardiogram (ECG)-based devices that are commercially available and will continuously monitor heart rhythm, will automatically identify arrhythmia, and allow ECG recordings to be made at the user's request. Users can capture irregular heart activity as they go about their daily lives or during extended monitoring periods. In addition, many large population studies (e.g., Fitbit Heart Study) have validated these findings on a population basis. There is also a high level of user

satisfaction (Lubitz et al., 2022). Specifically, users of PPG-based wearables are able to detect irregular rhythms and receive notification of potentially AF at a predictive accuracy rate of up to 98%, which is verified with a subsequent ECG patch study. These notifications serve as early-warning signs and encourage users to obtain telehealth services and diagnostic/tests, which allow for the identification of AF in many patients who may have been asymptomatic.

In addition to providing alerts, wearables can also give supportive outputs showing diagnostic results (eg, rhythm classification summaries, episode frequency, and clinician-reviewed ECG-confirmed reports). The Fitbit heart study demonstrated how the integration of alerts from wearables with telehealth workflows and mailed ECG patch monitors facilitated efficient follow-up without requiring in-person visits (Lubitz et al., 2022). Likewise, combining passive PPG monitoring with user-initiated ECG recordings increased diagnostic accuracy and decreased the number of false positives reported by physicians (Baman et al., 2022).

Additionally, wearables assist in clinical decision-making by identifying patients with a greater burden of AF; supporting referrals; and providing information to help clinicians make treatment decisions, including whether or not to use anticoagulant therapy or additional monitoring. Evidence from both the Apple Heart study and the Fitbit heart study indicates that detection of clinically significant AF using wearables allows for continued efficient provision of downstream care (Turakhia et al., 2019; Lubitz et al., 2022). In conclusion, mHealth wearables provide real-time rhythm alerts, ECG-derived data for confirmation of AF, longitudinal rhythm summaries, and telehealth-based reports, supporting early detection

of AF and improved clinical decision-making in resource-poor facilities and outside the hospital environment, where traditional means may not be available.

5. Performance and System Quality Evaluation using the ISO 25010 Model

Quantified based on the percentage of false positives generated during AF episode detection, as compared to the number of patients reported to the system. (Zhu et al., 2022; Cinotti et al., 2024). The functional correctness of the identification of AF episodes is supported by the percentage correlation of 98% between the data collected using both wearables and the clinical ground truth (Zhu et al., 2022). Performance efficiency was established using an Edge AI paradigm in conjunction with optimising the application of 1D-CNN, which ensured that computing times were comparable across devices. For example, the weighted time ratio (Cinotti et al., 2024; Plata et al., 2024) during classification of the signals was measured to be 0.100msec. Also, established a reliability measure based on the number of false positives raised by the system when detecting episodes of AF and the number of patients who reported episodes to the system. (Zhu et al., 2022; Cinotti et al., 2024). The functional correctness of the detection of episodes of AF has been corroborated by the 98% correlation between the wearable data collected and the clinical reference standard (Zhu et al., 2022). Access to private data for healthcare providers (da Costa & et al., 2024) or for patients (Egala & et al., 2021; Sumathi & et al., 2022) will require all users to work within the limits of their permissions to access those private records of their own. Additionally, the usability of the application (5.5) will also be evaluated based on the criteria

of satisfaction, usability, and experience to determine if the system's user interface provides an acceptable level of usability for both the non-specialist provider and the patient (Auton et al., 2023; Moorthy & et al., 2024). Finally, in order to ensure that the system remains maintainable and portable (5.6), the code quality metrics are managed to help determine if there is an acceptable level of complexity and coupling in the application and by the use of interoperability standards the system will be sustainable over time within a primary care workflow (Bertolino & et al., 2024; Klima & et al., 2022). ISO 25010 is used as a baseline for measuring quality, but it does not define such key features of IoT (internet of things), like trust and consumption of energy, which is leading the next version of this system to implement an edge processing model to accomplish its goal of meeting real-time responsiveness in response to use by providing clinicians with access to their clinical data in real-time (Cinotti & et al., 2024; Mehak & et al., 2024).

5.1 Identification of Relevant PPG Signal Inputs and Physiological Parameters for AF Detection

Research performed by Ding et al. (2024) indicates that large, standardized databases containing accurate and consistent PPG signals can yield highly accurate data for the identification of AF in a clinically viable manner. A database of waveforms containing PPG signals will provide the necessary data to analyze these signals for AF. This data will be initiated from the raw PPG waveform, using pulse peaks from the waveform as measurement parameters to compute IBIs for rhythm analysis (e.g., AF). Irregular and unpredictable pulse patterns will indicate the presence of AF. Typical physiological indicators of AF include an increased heart rate, pronounced beat-to-beat interval variability, and rhythm irregularity.

There are also observable waveforms, such as changes in waveform amplitude and shape, that are indicative of the presence of AF. Sufficient preprocessing steps must be taken prior to PPG signal analysis, including filtering and reducing motion artifacts; the resulting data after completion of preprocessing steps should help to assure high-quality PPG signal inputs for the purposes of identifying AF. Overall, this research shows PPG to be a viable method of monitoring the presence of AF using large-scale data sets and machine learning algorithms applied to millions of PPG measurements across real-world, ambulatory monitoring scenarios.

Photoplethysmography (PPG) signals can be utilized to monitor atrial fibrillation (AF) and atrial flutter (AFL). PPG is a method of monitoring heart rate through the identification of a raw PPG waveform that represents every heartbeat. The device uses a pulse peak from this waveform to compute the time span between heartbeats and identify irregularities in rhythm. The key physiological variables of AF and AFL are heart rate and the variation in the time interval between heartbeats and the irregularity of the heart's rhythm. AF and AFL can each be characterized as an irregularly changing heart pulse (i.e., the PPG waveform) and the shape of each heart pulse may help indicate the presence of AF and AFL, particularly when considering changes in the amplitude of the heart pulse(s). Of great importance to this analysis are the known effects of environmental noise and motion artifacts on the ability to accurately identify heartbeats and heart rate variability in PPG data; therefore, any time PPG data is analyzed for diagnosis, preprocessing steps are essential, such as the filtering of noise and assessing the quality of the data (Eerikainen et al., 2020). Finally, PPG is an effective tool for monitoring cardiovascular function non-invasively, as compared to the use of an electrocardiogram (ECG) signal, and validating AF and AFL will

depend on the ability to correlate the presence of AF and AFI as indicated via PPG signals with the ECG waveform.

Photoplethysmography (PPG) can identify heart rhythm problems such as atrial fibrillation with excellent accuracy. This process can be divided into three main steps. First, the peaks of the PPG waveform (or swingup) are used to identify the PPG rhythm by assigning an inter-beat interval (IBI) value and back-calculating the pulse to determine heart rate. Second, estimates of beat-to-beat variability and timing are made in order to detect problems with rhythm. Third, changes in the shape and amplitude of the PPG pulse will provide specific information about potential arrhythmias (eg, rhythms that are either too fast or too slow, or rhythms that are not sufficiently consistent). Beat-to-beat variability will also assist in determining the overall health of an individual's heart. PPG waveform characteristics can be distorted by excessive noise within the PPG signal resulting from both biological factors and environmental conditions. There can be some contamination of the PPG signal due to motion, causing motion artefacts and inconsistencies in the characteristics of the PPG waveform. A properly constructed PPG dataset can be used for frequency analysis and to aid in the diagnosis and assessment of arrhythmias.

Substantial research indicates that wearable optical sensors that continuously monitor peripheral blood volume play a critical role in AF identification through photoplethysmography (PPG). The PPG waveform provides the main signal input from which the peak of each pulse is extracted to derive interbeat intervals (IBI) for the analysis of rhythm. AF is characterized by an irregularly timed beat; therefore, measurement of heart rate, beat-to-beat interval variability, and rhythm irregularity will provide physiologic

measures that are relevant to the assessment of AF. Characteristics of the morphological waveform, such as alterations in amplitude and shape, will yield information that will assist in the identification of AF. For reliable and accurate identification of AF clinical implementation in out-of-hospital settings, preprocessing steps such as filtering, noise reduction, motion artifact reduction, and signal quality determination will be critical. Various PPG-based AF detection techniques differ primarily in their approaches to signal preprocessing, the extraction of features, and the modeling of extracted features. Modern techniques increasingly rely on deep learning to automatically extract temporal and morphological features from raw PPG data, while traditional methods tend to use engineered features (e.g., statistical measures of IBI variability and morphological features). This difference is important because models utilizing deep learning techniques provide better detection performance than traditional methods, even under noisy and/or ambulatory conditions, due to their ability to extract complex non-linear relationships between the dependent and independent variables.

Atrial fibrillation (AF) is an example of a cardiac rhythm disorder, which can be detected using photoplethysmography (PPG) signals collected from wearable devices. The process of using PPG to detect arrhythmia begins with obtaining a continuous, raw PPG signal. Pulse peaks are identified on the recorded PPG waveform, which enables researchers to compute intervals between heartbeats (IBIs) and to analyze how consistently the heart beats. Researchers can identify arrhythmias by identifying patterns of beat-to-beat irregularity in the heart rate and variability in beat-to-beat intervals. Since AF presents itself as non-regular intervals between beats, or a chaotic pattern of pulsations (i.e., a lot of confusion in timing), comparing the shape of the pulses, and amplitude and contour of the

recorded PPG waveforms can provide insight into the mechanism(s) (i.e., the "why") of AF. To provide this information, researchers will perform preprocessing (e.g., filtering to reduce noise, controlling for motion, ensuring that data is accurate and consistent with the everyday environment) before attempting to analyze the PPG data. Researchers can then apply deep convolutional neural networks (CNNs) to extract both timing and shape features directly from the raw PPG signal data. Ultimately, the ability of wearable devices to obtain accurate heart rhythm tracking from PPG signals in a non-invasive manner 24 hours per day will convey great improvements in both present-day healthcare and future healthcare practices (Bulut et al., 2025).

The wrist-worn smartwatches have the capability of detecting atrial fibrillation via continuous photoplethysmography (PPG) signals and better accuracy in detecting AF is achieved when longer recording times are used to obtain periods of beat to beat data. Primarily, raw PPG waveforms are the principal signal source, and pulse peaks are extracted from the PPG waveform to derive Interbeat intervals (IBIs) and to quantify irregularities in rhythm. Irregular pulse patterns and intervals are defining characteristics of AF and arrhythmias, and therefore beat to beat, heart rate, and irregularity indices are important indicators of the physiological status of the patient. In the event of adequate signal quality, morphological features of the waveform, such as changes in pulse shape, and amplitude, etc. can also aid in the detection of AF and arrhythmias. Signal preprocessing, such as filtering, motion artifact removal, and reduction of noise, is required, especially in an ambulatory setting, where movement and other rhythms may degrade the quality of the PPG signal. (Liao et al., 2022)

A change in your blood volume is detected through photoplethysmography (PPG) every time your heart beats, and the data is then used to select a raw waveform from this signal. The waveform allows the watch to determine when a pulse peak occurs and is able to calculate inter-beat intervals between beats which help identify rhythm irregularities. You will measure heart rate and the degree to which there are deviations of the time intervals between the beats and additional measurements to demonstrate how there is irregularity in the beating pattern. AF is a non-linear pattern and does not show a normal predictable beating pattern and is able to be assessed for abnormalities present with the PPG waveform, including changes in amplitude or shape of the waveform. There will also be some kind of wrist motion or noise that will need to be filtered, and artifact removed, or corrected in order to have a clean PPG signal coming from a wearable PPG device. If proper noise filtration can be accomplished, then it will be possible to monitor AF in a continuous and non-invasively and practically using PPG from a wearable device that a user can wear on their wrist (Bashar et al., 2019).

5.2 IoT-Enabled Photoplethysmography Systems and Machine Learning Techniques for Atrial Fibrillation Detection

IoT-enabled systems using photoplethysmography (PPG) have been created that can connect all the components necessary for wearables and their communication through machine learning, data analysis, and clinical follow up in one system for detection of atrial fibrillation (AF). The system is composed of a device that can measure heart rhythm worn on the body. Once data is collected in the device, it's transmitted, formatted in a standard way (e.g., as JSON) and delivered through a message broker to provide timely data transfer between components in the IoT-based health care system with minimal physical distance.

The data must be preprocessed before collection of the PPG to create clinically relevant features (inter-beat intervals (IBI), for example) needed for accurate detection of AF. Preprocessing includes techniques for removing noise, correcting for motion artifacts and identifying beats.

Machine learning to analyze heart rhythm patterns is well established as a powerful analysis technique in how researchers characterize rhythm generated from photoplethysmography (PPG) and aids in detecting the rhythm associated with the heartbeat. According to Cañón-Clavijo and co-workers (2023), there were 3 algorithms utilized for the evaluation of the PPG signals analysed (K-Nearest Neighbors [KNN], Convolutional Neural Network [CNN], Random Forest [RF]), where KNN had superior performance for classification accuracy and computational resource use. Specifically, KNN was the most appropriate algorithm to establish against newly created PPG with previously created PPG rhythms by using the time interval between two consecutive heartbeats, during low resource Internet of Things (IoT) environments. Moreover, Pereira et al. (2020) conducted similar KNN-based analysis and also determined that the variability of the heart rate from both PPG or Inter-Beat Interval (IBI) signals in measurements provide sufficient discriminative ability to subsequently apply supervised machine learning techniques using variability-based features and efficient computational algorithms to provide scalable, real-time monitoring of atrial fibrillation.

Wearable IoT-enabled PPG device effectiveness is published in the research of (S et al., 2020). Their study showed that ring-shaped wearable PPG devices like CardioTracker (CART) could collect high-quality PPG signals from the user's finger using high-intensity LEDs and photodiodes, allowing the user to receive continuous and convenient home-based

monitoring. They found finger-based PPG signals to have better quality than wrist-based PPG signals and utilize peak-to-peak interval variability analysis and irregular pulse rate measurements to improve AF detection reliability. All nine layers of the CNN integrated into the IoT system classify PPG signals automatically as either sinus or AF without utilizing any manually engineered features. Their performance was above traditional machine learning methods of AF detection (i.e. support vector machines) and provided high overall diagnostic accuracy, more so than traditional ML methods. This was further evidence of the effectiveness of applying deep learning techniques in conjunction with PPG devices for AF detection.

(Hiraoka et al., 2022) performed a study to evaluate the use of a PPG sensor-equipped Apple Watch for detecting AF at an early stage. They looked specifically at patients recovering from cardiac surgery. The authors utilized machine-learning algorithms to analyze the pulse rate collected from the wearable device in comparison to telemetry monitoring and 12-lead ECG. The results of their analysis showed that PPG pulse rates had strong associations with confirmed AF episodes from the ECG. Furthermore, the accuracy of the machine learning model used to achieve the results was high. The authors did state that while PPG rates and ECG rates did not always agree, the results of this study indicate that PPG-based wearable monitoring devices can be safely and feasibly used for early AF detection in the clinical setting.

The researchers' first step was to work out how to make a cost-effective Internet of Things (IoT) device that would record instances of atrial fibrillation (AF) by using classification of photoplethysmographic (PPG) data stream combined with machine learning (ML)/artificial intelligence (AI) on Sensor-Microcontroller pulses. As offered by Rhodes et

al. (2024), the authors presented an implementation of an inexpensive way of streaming electronic health record (EHR) data for AF classification via remote patient monitoring. Here the stream rate is set to capture only a single short (i.e. 15 seconds) PPG recording per patient. It was theorized that with an adequate number of stream recordings collected from a patient, the machine learning (ML) on the recorded PPG data would reach an AF qualified classification.

5.3 A Machine Learning Framework to Provide Interpretable and Edge-Optimized Real-Time Atrial Fibrillation Classification from Beat-Level Photoplethysmographic Data (PPG) in Wearable Devices

With the development of wearable devices has come an improvement in our ability to monitor one's cardiovascular health continuously, especially regarding atrial fibrillation (AF) — one of the most common cardiac arrhythmias on this planet and linked with high rates of stroke, heart failure, and a variety of serious cardiovascular illnesses (Mahim et al 2024). Detecting AF early can help reduce the risks listed above, and therefore machine learning (ML) methods were introduced around a decade ago as ways to analyze physiologic signals for real-time and accurate atrial fibrillation (AF) classification in PPG and electrocardiography (ECG). There are numerous studies that have been published on this subject, and while their goals include improving classification accuracy, computational speed, robustness to different external environmental factors, and the interpretive ability of the ML models, they use different methods on achieving these goals.

Wearable sensors have the potential to deliver reliable medical data in a reliable and valid form, however their physiological signals are subject to disturbances or “noise,” which

limits their ability to provide valid medical data (Walsh & McKee, 2022). Photoplethysmography (PPG) is one example of a clinically relevant, non-invasive wearable sensor that is frequently utilized in many medical settings; however, it is widely accepted that photoplethysmographs are often afflicted by motion-related disruption, drift and ambient noise (Almarshad, 2022) that will significantly impair the ML classification of atrial fibrillation (AF) via a PPG signal. Zheng et al. resolved this problem by creating Tiny-PPG; a deep neural network (DNN) with an extremely fast (small model size) runtime for real-time segmenting of artifactual PPG signals as they are recorded by edge devices. The Tiny-PPG used depthwise separable convolutions as well as a two-dimensional atrous spatial pyramid pooling module to minimize model size and maximize accuracy. Using 19,726 model parameters; an accuracy of 87.4% for detecting artifacts was achieved when all the PPG data were filtered with a baseline filter; 93.4% when using unfiltered baseline data. This study suggests that the development of artifact-aware preprocessing will enable the implementation of cleaning transformations for artifact detection and preprocessing of physiological signals in resource-constrained environments prior to performing ML classification on PPG signals.

Automation of predictive analytics in real-time on wearable devices necessitates the engineering of features as a key stage following data preprocessing. For instance, Zhang et al. (2021) demonstrated a proof-of-concept using a combination of classical methods for processing signals and machine learning (ML) to estimate HRV from a PPG signal. Error rates ranged from 3.5%-25.7%, and the models developed can be used for deployment onto embedded hardware as compact models. Through an analogous approach, Kumar et al. (2022) developed an ultra-edge arrhythmia detection pipeline with very high accuracy and lower memory requirements than traditional models, utilising effective feature engineering

techniques to enable small classifiers to achieve very high accuracy. These studies illustrate the benefit of extracting interpretable features from a PPG signal before embarking on predictive analytics in real-time on wearable devices, while also overcoming limitations of available hardware resources.

The creation of different types of representations for data in the area of ML-based AF detection is another very important area for development within this space. (Zhang et al., 2021) developed a Multiview deep learning approach that classified a single ECG signal using multiple representations derived from multimedia sources. The media used for classifying the ECG signal included time domain, spectrogram and Lorenz R-R scatter plots based on R-R intervals. The results of utilizing multiple representations simultaneously resulted in a more robust model with the potential to learn the temporal and structural characteristics of the cardiac signal. In a separate study (Bashar et al., 2020), researchers used the multimodal data provided by a smartwatch (i.e., PPG, accelerometer and heart rate) in order to improve AF detection. While both studies are attempting to reduce their false positives and create a more robust model, the methods used in the two studies are completely different—Zhang et al., (2021) employed a Multiview approach to classify one signal using many representations, whereas Bashar et al., (2020) employed a multimodal approach using different types of signals. The combination of both methodologies (Multiview + Multimodal) creates the potential to develop a ML system that is highly accurate and can execute tasks within seconds of being performed on a wearable electronic device.

Attention mechanisms have also been recognised as key to improving the performance and efficiency of machine learning (ML) models, along with the structure of the model. The authors Mäkynen et al. (2022) demonstrated that channel attention can be utilized

for the assessment of ECG and PPG signals. By using channel attention in an ML model, the model can take advantage of the information available in the most relevant elements of the data, whilst also reducing computation. Attention can automatically identify the portions of the ECG/PPG signal that contribute to the detection of atrial fibrillation (AF), whereas manual feature selection would require human intervention to identify the relevant portions of each signal. Integrating multiview and attention mechanisms into the development of ML algorithms for AF detection will provide the potential for high accuracy and interpretability of ML algorithms in real time on edge devices (e.g., wearables).

The clinical usage of machine-learning based tools to conduct AF detection will rely heavily on their interpretative capabilities. [Attia et al., 2020] conducted an experiment using ECGs and deep learning algorithms whereby morphological patterns (which were not seen by humans) associated with atrial fibrillation were able to be identified by different models of deep learning. This resulted in af detection through deep learning models being able to utilize certain features from the ECG that correspond to physiological differences. The results of this research indicate that neural networks can learn meaningful ECG characteristics that can enable accurate af detection while still being understandable to clinicians. Thus, by using attention based mechanisms in combination with these types of models, health care workers will be able to better understand the reason(s) why a given ECG segment was classified as AF thus providing an explanation as to how these machine learning predictions relate to the clinical reasoning processes of the health care practitioner. These types of interpretable models will be very useful for edge level deployments on either wearable devices or portable health monitoring solutions due to their ability to preserve transparency and computational efficiency.

One important consideration in the successful implementation of wearable technologies in real time is hardware and system-level optimization. (Akram et al., 2025) studied cardiac monitoring using the following technologies: TinyML, federated learning, quantized neural networks, and FPGA-based acceleration to identify accurate, privacy-preserving, and computationally constrained approaches. Additionally, (Mahim et al., 2024) developed architectures based on the ConvMixer-Transformer model that efficiently extract local and global features from ECG signals. Combining model optimization methods with interpretable Machine learning enables edge-level systems to conduct AF detection with high accuracy, low latency, and privacy. Thus, this combination has addressed the research question by demonstrating an interpretable Machine Learning method for classifying PPG-beat features in real-time on resource-constrained devices.

Many of the studies that have previously been conducted for AF detection fail to consider all the aspects involved; rather they analyze them individually from one another as independent features. Each of the main "features" of AF detection has typically received separate attention: artifact-aware preprocessing, feature extraction, feature enhancement (refined feature enhancement) using an attention-guided approach (using only a small subset of available measurements), multiview/multimodal integration of extracted features and their respective explainability (if applicable), and finally, feature deployment in a hardware-aware manner. The inconsistency of different studies that do not integrate all of these different steps makes it much more difficult for researchers to develop a machine-learning (ML) model that is interpretable and suitable for real-time edge wearable monitoring of AF. An integrated and unified framework, which combines these four different pipelines, will ultimately make it possible to create a practical, efficient ML framework to enable isolated researchers from

around the world to help provide accurate and reliable systems to detect AF via machine-learning techniques.

To overcome this deficiency, this paper introduces a unified framework for implementing machine learning systems for AF detection purposes. The proposed unified framework consists of four discrete pipelines, each building upon the previous stage: (1) an artifact-aware preprocessing pipeline executed at the network edge; (2) a hybrid feature extraction (HFE) module combining statistical inter-beat interval descriptors with morphological waveform features; (3) an attention-guided, interpretable classification layer using a lightweight 1D-CNN architecture optimized for edge deployment; and (4) a multiview output module that generates confidence-weighted alerts, rhythm irregularity summaries, and clinician-facing decision-support indicators.

5.4 Development of System Outputs and Decision-Support Functionalities for Early Atrial Fibrillation Detection

Recent innovations within mobile health technology have changed how atrial fibrillation studies focus and assess atrial fibrillation. Originally, the focus was primarily on simply measuring detection accuracy, but as wearable tracker technologies evolve, there is increasing effort on making those systems' outputs as clinically applicable as possible. Most of the recent atrial fibrillation validation studies examined the various sensing technologies employed (e.g., single-lead electrocardiogram [ECG] vs. photoplethysmography) and their ability to measure the presence of an atrial fibrillation-related signal, while more recent studies have included how the digital outputs from the detection algorithms are transmitted, used, and integrated with existing healthcare delivery systems.

Large-scale screening trials have created a basic notification-based output structure for wearable technologies. The Apple Heart Study pilot project led by Dr. Mintu Turakhia was able to validate the use of an irregular pulse detection with a smartwatch across a sample of over 400,000 participants. For those participants who met set criteria for a detected irregular pulse, an event notification would be generated with instructions to undergo further evaluation via ECG patch (Turakhia, et al., 2019). The binary algorithm used to create the notification output (i.e., irregular vs. non-irregular) is designed to deliver simple, straightforward communication to end-users when detecting an irregular pulse and promote widespread implementation as compared with requiring a detailed analysis/interpretation of the ECG result for clinical decision making. Therefore, the primary intent of the wearable trackers was to provide clinical evidence for screening and not be an all-inclusive clinical decision-making tool.

In addition to wearable technology for consumers, integrated telehealth systems have better organized output capabilities. One example of this is the TeleCheck-AF project which provided application-based rhythm monitoring during the COVID pandemic, and enabled patients to record their own rhythm data to be directly embedded in workflows used during teleconsultation (Pluymaekers et al., 2021). The TeleCheck-AF system was capable of transmitting structured rhythm summaries to clinicians prior to their consultations with patients, thereby allowing for a rhythm-based therapeutic decision-making process. System output from the TeleCheck-AF project were integrated into clinical processes, thus converting detection information to decision support tools for actionable clinical purposes. The value of output formatting was also highlighted in studies conducted as part of clinical validation studies. An example of this is the FibriCheck validation study which compared

mobile phone-based PPG detection with the gold standard reference ECG, and reported high levels of sensitivity and very high levels of specificity; (Proesmans et al., 2019). More importantly, the mobile application produced rhythm reports that were actionable by physicians, and included indications of episode length and the quality of the measurements taken from the PPG. The incorporation of signal quality information is a significant enhancement to decision-making by providing the decision-maker with a means of judging the reliability of the assessment that was conducted prior to the physician acting; whereas binary notification systems do not provide sufficient context for the recipient to determine the quality of the information received in their notification.

Research using randomized clinical trial data demonstrates how monitoring different variables can influence treatment decisions. Continuous ECG patch monitoring of atrial fibrillation (AF) using telemonitoring has been shown to improve AF detection and the initiation of anticoagulation therapy compared to usual care (Steinhubl et al., 2018). The mSToPS study used telemonitoring of AF through continuous ECG recording rather than alerting via the smart watch technology available at that time; therefore, the ECG recordings from the mSToPS study had been evaluated clinically and produced data directly affecting pharmacologic management. Thus, evidence demonstrates that how clinicians use structured output data from verified automated systems can lead to significant changes in providing healthcare to patients. Similarly, outputs from AI-based classification systems allow for probabilistic, automated outputs that are more than simply event triggered alerts. For example, deep neural networks learn to recognize AF using single-lead ECG recordings; however, studies have also raised concerns about how interpretable these outputs are to clinicians (Hannun et al., 2019). Thus, while probabilistic outputs allow for more

sophisticated diagnostic assessments, they do not provide clinicians with an easy interpretation of what to do with the data provided.

Ultimately, both the quality of the AF detection systems' ability to provide accurate information on AF, as well as, the quality of how that information is presented to clinicians will determine how well they will be able to effectively utilize the AF detection devices. The trend from previous studies clearly illustrates a common pattern of progression from just providing clinicians a simple notification of an event that can lead to AF, to now providing a much longer term structured and supportive decision system for healthcare providers. Future design efforts for AF detection devices should focus heavily on providing interpretability of the detection/notification systems, developing a robust risk stratification system and integrating the technology into established workflows of current health care services.

5.5 Performance and System Quality Evaluation using the ISO 25010 Model

The ISO/IEC 25010 model is an essential part of the ISO/IEC 25000 (SQuaRE) framework and provides a structured method of evaluating software product quality and resulting user experience (Souza-Pereira et al., 2021). According to Argotti et al. (2024), an effective management evaluation system for medical Internet of Things (IoT) requires a comprehensive approach that balances both system efficiency and clinical reliability. Traditionally software assessments have placed more emphasis on evaluating code quality or the performance of embedded systems. The ISO/IEC 25010 model has eight (8) primary quality characteristics; this study will evaluate six (6) of the characteristics (Functional Suitability, Performance Efficiency, Reliability, Security, Usability, Maintainability) that are further broken into thirty-one (31) supporting sub-characteristics. In addition to performing

as a checklist for evaluating embedded technologies in cardiac monitoring, the ISO/IEC 25010 model provides a practical framework for understanding how embedded technologies perform when constrained by real world medical conditions. Argotti et al. (2024) further explicate that unlike traditional computer systems, there is no current standard for evaluating embedded systems/software under domain-specific constraints of safety, security, limited resources and real-time performance. Therefore, it is essential to build upon these domain-specific quality constraints throughout the quality model development process for embedded software to ensure consistent application of these constraints throughout the development cycle.

Functional Suitability describes how accurately the system generates the right results and the level of precision necessary for those results(s). This is often the primary means by which a system's ability to perform its essential functions (or core capabilities) can be validated in a clinical setting.

Functionality is broken down into three categories based on ISO/IEC 25010 standards: functional completeness, functional correctness, and functional appropriateness.

The primary means for assessing the functional suitability of the system is through its ability to provide accurate detection of Atrial Fibrillation (AF) and to provide the necessary diagnostic support through (Tran et al., 2023; Zhu et al., 2022). These studies reported that PPG based wearables have demonstrated the ability to have high sensitivity and specificity in detecting arrhythmias through PPG algorithms. As reported by Zhu et al., (2022), PPG based wearables have shown to have robust, consistent performance with 87.8% sensitivity and 97.4% specificity in detecting arrhythmias at the same time that Tran et al., (2023) reported that the PPG based smart wearables demonstrate mean 91% accuracy in predicting AF and

100% sensitivity/99% specificity in the detection of persistent AF when using readily available consumer devices (such as smartwatches).

The strong functional correctness is critical to the timely identification of asymptomatic Atrial Fibrillation (AF), as asymptomatic AF, if not treated promptly, puts patients at a significant risk of developing complications, such as stroke. In addition to these results, modern system designs can further improve reliability through the use of advanced algorithmic blocks that provide adequate coverage of the data and estimation of accurate values

In the context of the ISO 25010 performance efficiency, the efficiency of a product with regard to its capabilities are the functions it offers, to meet stated requirements under specified conditions, as well as the cost. Therefore, an IoT framework designed for the detection of cardiac arrhythmia must ensure that transitioning from cloud-based analytics to edge processing will not negatively affect the clinical monitoring real-time requirements. According to Cinotti et al. (2025), the main metric used to access the efficiency of the edge processing is the latency of signal classification. The results of the prototype implementation. Proving that the system sustains low-latency and deterministic processing. The associated computational workload of the core basis of the AI computing system (1D-CNN) used to determine the classification of heart rhythm can be performed on the target micro-controller in 2.7 ms. While the actual time needed for inference processing is 23.16 ms when the timing associated with data management and pre-processing functions are included. Classification events occur every 78.27 seconds which translates to a 100 RR-interval window; therefore, the system is able to remain in low-power idle mode for the majority of its monitoring duration. Also, acquisition routines triggered by the ECG front-end occur every 255.99 ms

and only last for 2.49 ms; thus, causing no significant impact to concurrent background processes or applications. (as indicated in Sabbadini et al. 2022) The computational efficiency of the framework results from the NN structure and is created primarily as a result of the NN design; with the NN composed of 4 hidden layers containing 5, 30, 20, and 10 nodes.

This lightweight system is designed to use the limited computing capability of the wearable sensors it is deployed on while producing a classification of Atrial Fibrillation (AF) with an F1 score of 0.94. This high level of diagnostic accuracy from a low complexity classification model is important to enable the system to operate within the limited resources available for a non-embedded device. One of the performance metrics for this model is its capability to perform offline inference. When used on a wearable device, the classification speed can maintain the same level with offline inference as it could if the device were connected to the internet and had to communicate between devices via two-way communications in order to complete a cloud-based classification.

In the context of the ISO/IEC22410 Standard (ISO/IEC), reliability is determined by how well the system is able to execute its expected functionality while being exposed to environmental conditions over a defined timeframe. In the case of a device for detecting arrhythmias using PPG data, reliability will depend upon the maturity of the signal acquisition technology, the ability to tolerate environmental noise, and check that diagnostic information derived from PPG data is accurate. As reported by Lyimo and Ngulugulu (2025), the hardware system is recognised as having reached a level of maturity when a transmission configuration consisting of red (660 nm) and infrared (940 nm) LEDs is used. This hardware configuration provides both stable and consistent acquisitions of total body surface area

(TBS) signals that show a high degree of correlation with commercial-grade pulse oximeters. Thus, derived TBS data would be stable and therefore reproducible even in low-resource conditions. However, fault tolerance remains one of the major issues affecting the reliability of wearables. In a study conducted by Filgueiras and others (2025), it was shown that while cheap sensors can deliver precise readings such as heart rate and SpO2 levels, they have limited reliability based on the user's physical conditions; e.g., whether the individual is resting, sitting or standing position. Therefore, in order to achieve reliable output, an Internet-of-Things framework must take anatomical and postural characteristics into account when designing systems that mitigate motion artifacts which could result in inaccurate septal arrhythmia classifications. The diagnostic reliability and availability of the device is also supported by comparing its results to established clinical “gold standards” (Tran et al., 2023). Furthermore, Tran et al. (2023) suggest that wearable health devices (WHDs) have reached enough maturity to allow them to successfully identify several types of cardiac arrhythmias in an outpatient environment with ongoing monitoring, which intermittent ECGs are typically unable to fulfil. Finally, while PPG-based smartwatches offer greater availability than traditional ECG monitoring devices for providing long-term continuous screening for AF, Sibomana and co-researchers (2025) note that ECG smart chest patches will give users slightly higher reliability because the gap between ECG and PPG is large enough to highlight the need for developing improved signal processing methods in order to ensure this system is acceptable in everyday practice. The use of blockchain technology in conjunction with security with the aim of maintaining an independent and secure ecosystem has been discussed by Sumathi et al. (2022). Healthcare data is stored securely and managed through controlled access to databases stored on the Internet of Things (IoT) environments with the

use of blockchain technology. In this case, the use of a decentralized ledger prevents unauthorized modifications to physiological data, therefore making it tamper-proof and verifiable. Therefore, the security of this system is improved considerably through the use of a decentralized system versus a centralized system because centralized systems have a lower incidence of unauthorized access and can be severely compromised through single-point-of-failure attacks or data breaches. The security of these protocols is further enhanced through the use of privacy-enhancing technologies that allow for controlled sharing of medical records across different cloud servers. Egala et al. (2021) have shown that blockchain-based systems are able to facilitate the secure sharing of health-related IoT data while also protecting the anonymity of users and the encryption of data. Based on their findings, they conclude that blockchains can automate the authorization process for the decryption and viewing of a patient's cardiac data, ensuring that only those individuals who have permission can view this information. As a result, this provides the infrastructure necessary to maintain a high level of security and privacy for sensitive cardiac data without relying upon a central intermediary. To ensure patient's privacy/security while undertaking remote monitoring, it is essential to establish secure protocols. The organization's evaluation determined that because blockchain technology has properties that make records permanent and can verify an event's occurrence, there is a possibility to securely verify each occurrence of a detected risk for example Atrial Fibrillation using blockchain technology. A secure combination of these properties will provide a secure source of information regarding the medical condition of patients which will comply with confidentiality laws and therefore signify to use this method as a viable resource for long term cardiac management.

The definition of usability can be described as how well a specific product or system meets the requirements of designated users to achieve predetermined objectives, in a given environment, with minimum effort, with optimum performance and in a manner that is enjoyable. In the case of PPG-based arrhythmia detection systems being used in a non-hospital environment, usability will also depend on the ability to integrate the wearability of hardware with the ability to utilize software. The first factor affecting usability is the design of the hardware for wearability. For example, Guarducci et al. (2025) state that the ability of non-contact and wearable health devices to be effective is dependent on their inconspicuous nature. As cardiac patients utilize non-hospital settings, the device should be designed to allow the continuous monitoring of cardiac activity while also allowing the user to perform their typical daily activities. Lingawi (2025) further supports this concept by indicating that usability is achieved when real-time analysis algorithms provide the ability to mitigate the effects of motion on the accuracy of vital signs during normal activity. This will allow patients to have accurate assessments of vital sign parameters while not having to have an advanced level of technical competence to ensure that the integrity of the signal is maintained in a clinical setting. Lastly, the effectiveness of the software user interface for both patient users and non-patient users will also have a major impact on the overall usability of the system.

Community Nurses are the individuals who interpret data; their analysis, which is based on a combination of quantitative and qualitative methods, indicates that an appropriate interface will provide a balanced experience for both patient and nurse user. In other words, the patient will have access to simple data entry and the nurse will have the ability to perform a meaningful semantic analysis of the patient's status. This makes the satisfaction of the

system dependent on its capacity to provide timely, easy-to-understand feedback in a manner that allows for autonomous action.

The transition from raw PPG waveforms to clinical alerts which result in action has made the device usable for non-health professionals. The ability for real-time on-device inference decreases the cognitive burden of the user and affords him or her the opportunity to provide timely, meaningful intervention and thereby increased engagement in an individual's cardiac health management without requiring ongoing specialist monitoring.

Under the ISO/IEC 25010 standard, the long-term success of IoT frameworks for cardiac monitoring that incorporate machine learning is supported by the quality characteristics of Maintainability and Portability. These two essential pillars are critical to ensuring that the system will evolve to meet the changing needs of clinical practice, while remaining functional across the many different infrastructures of primary care workflows.

To ensure a continuous cardiac monitoring system has a long life span, the system should be designed for easy maintenance. Klima et al. (2022) have described how to quantify the maintainability of IoT systems using metrics including (but not limited to) analyzability, modifiability and testability. Because of the hardware limitations of wearable sensors, an IoT maintainable architecture needs to be modular in order to support iterative updating of machine learning models (e.g., from simple pulse detection to a specific arrhythmia classification) without having to rebuild the entire system. This modular design will facilitate the corrections to algorithms used for accuracy and to adjusting to recent cardiac research at minimal cost, resulting in the system remaining a trusted clinical tool for years to come.

In tandem, the portability of the IoT system will allow for the transference of the patient's data to multiple healthcare settings without issue. Da Costa et al. (2024) argue that the future of precision health will be the combination of real-time IoT data streaming with integrated digital healthcare records to create distributed, interoperable health histories; therefore, the system must be designed for high portability in order to address the various challenges created by multiple data sources that differ from one another, both technically and legally and with respect to regulations. The integration of interoperable standards and distributed ledger technology in order to ensure data integrity and traceability will allow for the myriad of datasets integrated to become a valuable resource for machine learning algorithms as they move between a patient's home environment and institutional primary care workflows.

In conclusion, the various aspects of this type of architecture create a data-sharing environment that facilitates the sharing of information in a sustainable and cooperative manner. Nature has provided us with an effective means to manage large amounts of information through using machine learning models to monitor and diagnose arrhythmias; these are best implemented in a portable and maintainable manner (Da Costa et al., 2024). The combination of these features will provide the basis for a more comprehensive health record for an arrhythmia detection system based on the Internet of Things that is not only secure and accurate but also has the technical capability to support the delivery of personalised treatments for individual patients and optimised clinical trials for prolonged periods of time (Klima et al., 2022; Da Costa et al., 2024).

Synthesis

The first step to producing a waveform that can identify atrial fibrillation (AF) from unfiltered photo-plethysmographic (PPG) data collected from wearable devices to identify and measure the peaks of the heart's normal beats (pulse peaks) and the time between those beats (inter-beat interval or IBI). The removal of noise, removal of motion-related artefacts and assessment of signal quality will all help identify pulse peaks and IBIs, and therefore generate accurate measurement features for identification of AF both in laboratory settings and under uncontrolled conditions. By combining these measurements with deep learning (DL) methods, specifically the 1D CNN Deep Learning Model, this will increase the likelihood of correct detection of AF without a need for invasive procedures. Also, due to the inherent properties of the human body, and the combined effectiveness of DL, AF can be accurately detected using PPG data of any type and any level of accuracy through multiple methods of identification of AF.

PPG, which stands for photoplethysmography, has been demonstrated to be similar between many different studies regarding detection of atrial fibrillation (AF) through the combined application of both Internet of Things (IoT) technology and deep learning. Each study has highlighted the same four components as being necessary for any study involving PPG and AF based on IoT and machine learning: Continuous PPG sensor wearability due to privacy and convenience reasons, Efficient transmission of data to a device on an ongoing basis, Comprehensive process for acquiring PPG signals prior to identifying AF events and, Use of deep learning (DL) algorithms to identify AF based on irregularities in pulse dynamics. According to Pereira et al. in 2020 and other sources, using inter-beat interval analysis (IBIA) and analyzing subsegment signal sections can greatly improve the efficiency of data transmission, reducing energy usage without affecting diagnostic accuracy, if both

methods do not cause a change in the quality of the algorithm to detect AF. Overall, the combined use of IoT data from PPG sensors with DL techniques will allow for a scalable method of detecting AF, remotely, with minimal invasiveness. Patients will thus have more options for obtaining high "quality" monitoring of AF than previously existed (i.e., via conventional ECGs).

This research paper presents a predictive and optimal machine learning model that classifies beat-level PPG signals collected using wearable technology in real time to detect AF (Atrial Fibrillation) using wearable devices. In the past few years, there have been many design principles and use cases associated with the detection of AF using wearable devices. Some examples include pre-processing of the signal to reduce the impact of motion and noise, extracting relevant low-complexity features, using attention mechanisms to guide model processing & ultimately, modeling data using a Multiview or multimodal representation will create greater resiliency against errors. Model interpretability has continued to gain significance, leading to the conclusion that models for wearables will be limited by power and processing restrictions. Most of the current methodologies that increase the efficacy of methods mentioned do so on an individual basis versus providing an end-to-end solution. Examples are studies that reduce artifacts, those that develop new attention-based networks, and those that design models for embedded systems (hardware). An area that is not well-addressed is the development of a unified framework that can combine these different components together to constitute a single system capable of detecting AF in real-time, in an interpretable manner, and on the edge. Developing a unified framework is of critical importance to future advances in the continuous monitoring of cardiovascular conditions using wearable technology, particularly when the use of machine

learning models provides not only accurate predictions of what will happen next but also predictions that are transparent, reliable, and clinically relevant in everyday practice.

The synthesis of the Development of System Outputs and Decision-Support Functionalities for Early Atrial Fibrillation Detection expose a significant gap in research. Evidence exists in high-income countries that wearable devices and AI-enabled systems can effectively detect AF, yet there is insufficient evidence to demonstrate the efficacy of such technologies at an individual level in low-resource countries or otherwise as they currently exist are both costly and difficult to obtain or use. Furthermore, several studies have been performed using edge-level processing to minimise the need for constant internet connectivity, and clinically actionable outputs and real-time visualization. Additionally, although considerable effort has been expended to develop population-level datasets to be used to integrate into routine primary care processes, relatively little has been accomplished thus far, particularly in LMIC countries.

These gaps provide justification for the current study, which proposes a novel IoT solution using the principles of machine learning to detect cardiac arrhythmias through a photoplethysmography-based sensor. The proposed system will provide affordable, clinically actionable outputs, edge-enabled processing, and early risk detection to underserved populations and address both of the technical limitations and contextual inequity documented in the literature by providing support for early atrial fibrillation detection and timely notification of users with integrated clinical decision support in non-hospital settings. While many studies demonstrate strong technical feasibility of detecting atrial fibrillation, many fewer provide extensive discussion regarding how detected events are communicated, contextualized and integrated into clinical decision-making. This lack of communication

regarding detected atrial fibrillation events is especially apparent when comparing population-level screening studies and integrated clinical decision support systems.

To effectively implement deep learning in medical diagnosis requires solutions for the inherent "black-box" nature of neural networks so that Explainable Artificial Intelligence (XAI) will play an important role in establishing clinical trust and transparency in connection with the use of deep learning in cardiovascular monitoring. As an example of the effectiveness of Grad-CAM, Jahmunah et al. (2022) have reported using Grad-CAM to provide explanations for deep learning model predictions for ECG signals, which gave clinicians the opportunity to visualize the detection of acute myocardial infarction. With the need for interpretable cardiac care addressed, new research conducted by Luo et al. (2025) has demonstrated that Grad-CAMs can accurately identify the specific morphological features influencing the classification of atrial fibrillation, namely, the absence of P-waves, elevated T-waves and irregular R-R intervals. Collectively, these studies demonstrate the value of XAI to connect the complex outputs from algorithms to the well-established diagnostic criteria used in clinical practice.

The project will be developed using the ISO/IEC 25010 framework providing an integrated approach to assess the viability of IoT Heart Monitoring systems from both a Clinical and Technical perspective, given the limitations of resource poor environments. By balancing the functional suitability of highly accurate arrhythmia detection with performance efficiency via low-latency edge processing, real-time diagnostic accuracy will not necessitate the use of high-complexity models or always-on internet connections. In addition, the technical underpinnings of the project will support the secure and private storage of sensitive medical data through the use of blockchain for decentralised security, allowing the hardware

to be reliable and keeping signal quality for heart rhythm analysis consistent despite variations in a patient's anatomical characteristics or physical condition. To achieve long-term, scalable integration of these frameworks into primary care workflows, emphasis must be placed on user-centred design so that both the patient-worn device and the user interface for patients and non-specialist care providers remain accessible and understandable. As the hardware and software systems will be developed using a maintainable modular architecture and built upon interoperable standards, the systems will evolve from being isolated technical prototypes to being complete clinical decision support systems. This research has proposed an alternate approach for developing a new framework for the collection of photoplethysmography (PPG) data and deep learning (DL) data that will provide actionable, interpretable, reliable, straightforward, and accurate data to consumers. The new framework will not only produce a binary signal to facilitate adequate detection, but provide consumers with the following types of data: 1) early warning signals, 2) rhythm irregularity trend data and 3) confidence weighted alerts. Each of these types of data may support time-sensitive alerts that can assist the consumer in determining an appropriate follow-up action without unnecessary anxiety. The summary monitoring reports provided to the medical care provider may also enhance the medical care provider's clinical decision-making by highlighting areas of risk and changes over time in rhythm behaviour. The framework for this study is designed with attention to affordability, clinical utility and edge level analytics. Furthermore, it is based on the fact that many of these devices are developed and tested in resource limited, and non-hospital settings. In addition, by addressing how the outputs/ functions of the devices can facilitate early intervention, patient engagement, and informed clinical response through the use of empirical data from existing

research; the framework attempts to bridge the gap between new advanced arrhythmia detection research, and the practical application (usage) of this research. Consequently, the study provides both a solution to the ongoing problem of under-diagnosing atrial fibrillation (AF) and other forms of arrhythmias; and also a context to support the solution. As such, this study represents an important contribution to the literature concerning AF detection.

Much research has demonstrated that mobile/wearable/IoT devices can detect AF; however, there is also a significant research gap that exists regarding designing these devices so that the detection can then lead to effective early management of the disease outside of a clinical/hospital setting (called the ‘outpatient setting’). This is particularly important for those individuals with AF that may not have the opportunity to visit a clinic or hospital. The primary obstacles inhibiting the effectiveness of currently available devices include the low interpretability of the output of existing devices or algorithms; the high cost of infrastructure needed for existing devices; and a lack of adequate integration into the workflow of primary care providers (PCP). Our proposed research will assist in addressing these barriers.

Conceptual Framework

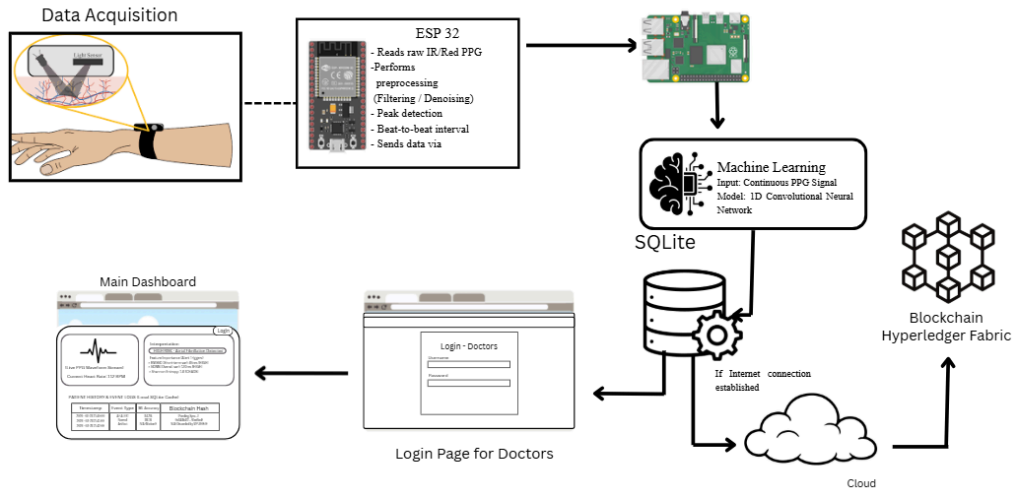


Figure 2.1 *Conceptual Paradigm*

Figure 2.1 illustrates the conceptual framework of the proposed IoT-based arrhythmia detection system integrating photoplethysmography sensing, edge preprocessing, machine learning classification, and blockchain-based secure data management. The framework presents a sequential and interconnected flow of processes beginning with physiological signal acquisition and culminating in secure event recording and visualization.

The process begins with the MAX30102 photoplethysmography sensor, which captures real-time cardiovascular activity by emitting infrared and red light into the user's skin and measuring variations in reflected light intensity caused by blood volume changes. These variations are converted into digital raw PPG signals and transmitted via I2C communication to the ESP32 microcontroller. The raw PPG waveform serves as the primary input of the system.

Upon acquisition, the ESP32-C3 performs preliminary signal conditioning and transmits the raw PPG data to the Raspberry Pi. The Raspberry Pi then functions as the primary preprocessing unit, applying bandpass filtering and motion artifact removal to the received signal. A peak detection algorithm is applied to identify systolic peaks, from which inter-beat intervals (IBI) are derived to support BPM computation and rhythm analysis.

The preprocessed and segmented PPG signal is transmitted to the Raspberry Pi, which operates as the deep learning processing unit. Rather than relying on manually engineered HRV features, the segmented PPG waveform is fed directly into a pre-trained 1D Convolutional Neural Network (1D-CNN) that has been trained offline on the MIMIC PERform AF Dataset (Charlton et al., 2022) prior to deployment. The 1D-CNN automatically extracts temporal and morphological features from the sequential signal data through its convolutional layers, eliminating the need for manual feature engineering. The model outputs a binary classification indicating whether the detected rhythm is normal or indicative of atrial fibrillation.

A decision node follows the classification stage. If no arrhythmia is detected, the system continues continuous monitoring and visualization without initiating secure recording. If arrhythmia is detected, the framework triggers the secure data management layer. In the secure data management layer, the arrhythmia classification result is combined with metadata including timestamp, device identifier, and computed features. A SHA-256 cryptographic hash is generated to ensure data integrity. The hash, event metadata, and raw PPG waveform data are stored locally in a SQLite database on the Raspberry Pi, ensuring continuous offline operation without requiring internet connectivity. Once an internet

connection is established, all locally cached records are synced to the Hyperledger Fabric blockchain network, ensuring immutability, traceability, and tamper resistance.

Simultaneously, the system provides real-time visualization of heart rate, PPG waveform, and arrhythmia detection status through a user interface. This ensures usability and supports clinical interpretation while maintaining continuous data logging for analysis and auditing. Overall, the figure demonstrates the integration of physiological sensing, edge computing, machine learning, and blockchain security into a unified architecture designed to provide real-time, secure, and reliable arrhythmia detection using photoplethysmography signals.

Definitions of Operational Terms

Arrhythmias.

Arrhythmias are detected by analyzing temporal and morphological patterns in the preprocessed PPG waveform using a 1D-CNN classifier, which automatically identifies deviations from normal sinus rhythm without relying on manually computed inter-beat interval features

Atrial Fibrillation (AF).

AF is a sustained irregular heart rhythm characterized by inconsistent and irregular atrial contractions and irregular pulse intervals. AF will be identified when the extracted PPG beat features are processed and classified by the developed 1D Convolutional Neural Network (1D-CNN) model. The model will output a binary classification indicating whether the patient has AF (1) or does not have AF (0).

Blockchain-Enhanced Security.

Blockchains are a type of decentralized data logging mechanism used to ensure that stored data is secure and can be trusted. In this research project, AF detection events will be hashed with SHA-256 and recorded to protect against the occurrence of modifications.

Cardiovascular Disease (CVD).

The term cardiovascular disease encompasses all diseases of heart and blood vessels and complications from cardiovascular disease include Stroke, Heart Failure and Sudden Cardiac Death.

Convolutional Neural Network (CNN).

A Convolutional Neural Network is a deep learning architecture designed to automatically extract spatial and temporal features from input data. The CNN component of the 1D-CNN model will process PPG signal data to learn distinguishing patterns associated with normal and abnormal heart rhythms.

1D-CNN (Intelligent Detection Using Convolutional Neural Networks).

1D-CNN is the proposed system that utilizes convolutional neural networks to automatically detect atrial fibrillation from PPG signals. It integrates signal processing, deep learning classification, and real-time alert mechanisms for continuous heart rhythm monitoring.

Early Detection.

Early detection of abnormal heart rhythm patterns refers to identifying those patterns before significant complications develop. Operationally, this is done by the system classifying AF in real-time and producing alerts that trigger timely medical consultations.

Edge-Level Processing.

This refers to the computational analysis taking place close to where data was generated, rather than relying solely on expensive, remote cloud servers to store and perform computations. The calculations performed in regard to this study will be the local filtering of the signal, the extraction of features and the classification of whether or not there is Atrial Fibrillation through the use of Embedded Devices.

Interpretable Machine Learning.

A 1D-CNN is utilized for classification as it is capable of automatically learning temporal and morphological features from sequential physiological signals such as PPG data. This approach enables more robust pattern recognition compared to traditional methods, as the model can capture complex, non-linear relationships within the signal without relying on manually defined thresholds.

Internet of Things (IoT).

The Internet of Things is an emerging collection of hardware and software devices connected to the internet that provide a means to capture transmitted or processed data. The

IoT component of this proposal will provide examples that illustrate how a wearable sensor, microcontroller unit, Raspberry Pi, database and web interface combine to provide physiological ongoing real-time monitoring of cardiac rhythm.

ISO 25010 Software Quality Model.

The ISO 25010 Software Quality Model is an organized framework used to evaluate how well a software system performs overall. Throughout this project, it will be utilized as an assessment tool in respect of the following criteria: Functional Suitability, Performance Efficiency, Reliability, Security, Usability, Maintainability, and Portability (of the manufactured product).

Low-Resource Environments.

Low-resource environments refer to communities or areas that possess little-to-no access to advanced medical resources and/or technical specialties. Within this study we are characterizing low resource environments in relation to affordable, portable and edge-compute enabled monitoring solutions needed to access AF screening.

Machine Learning.

An area of study that utilizes computational algorithms to analyze data and make predictions based on that analyzed data. In this particular study, there will be the analysis of

extracted PPG features (Photoplethysmography), utilizing more classical supervised learning techniques to classify heart rhythm as being in or not being in Atrial Fibrillation.

Non-Hospital Environments.

Non-hospital environments are any of the places or spaces outside of formally established health care facilities. As defined by operational level, non-hospital environments include homes, community barangay health offices, as well the general public wherever there exists an enabling environment.

Photoplethysmography (PPG).

Photoplethysmography is a non-invasive optical technique used to detect and measure changes in blood volume due to variations in light absorption by the blood. PPG data collected from a wrist-based device allows us to identify physiological signals associated with heartbeat detection and diagnosis of atrial fibrillation.

PPG-Beats Toolbox.

The PPG-Beats Toolbox is a platform for processing raw PPG waveform data using signal processing techniques in preparation for training machine learning algorithms. This project will use the PPG-Beats Toolbox in three steps: to pre-process PPG waveform data, identify PPG beats corresponding to each heartbeat (when blood volume increases) and create features that can be used to calculate time intervals between beats.

Real-Time Alerts.

Real-time alerts are system-generated notifications that are triggered by an abnormal heart rhythm detection. In the research project, when an AF episode has been classified, an alert is generated that is displayed on the OLED screen and dashboard interface.

Wearable Sensor.

A small device that is worn on the body to continuously capture physiological signal data. In this research project, a wrist-based PPG-enabled device will be used to obtain heart rhythm data for real time monitoring and classification.

CHAPTER 3

RESEARCH METHODOLOGY

This chapter contains the blueprint of the study, providing a structured and systematic framework concerning the methodology implemented in the study. This section allows the researchers to present the methods of the study to effectuate the answers to the problem.

Research Locale

The proposed study will be conducted in Caloocan City, a highly urbanized city located in the northern part of Metro Manila, Philippines. Caloocan was selected as the research locale due to its high population density and diverse communities, which provide a representative sample of urban residents for evaluating health monitoring technologies. The large population ensures access to participants from varying ages, socioeconomic backgrounds, and levels of health awareness, which is essential for gathering meaningful feedback on the proposed IoT-based cardiac arrhythmia detection system.

In addition, certain areas of Caloocan experience limited access to barangay health services, local clinics, or community healthcare programs, highlighting the need for alternative health-monitoring solutions. This context makes Caloocan an appropriate setting to assess the potential usefulness and relevance of a wearable cardiac monitoring system that operates independently of traditional healthcare facilities. Residents' perceptions in such areas are expected to provide valuable insights into the practicality and potential adoption of the system.

The city's urban environment and geographic diversity, including the North District (residential and industrial areas) and the South District (commercial centers), presents a realistic backdrop for studying the integration of wearable health technology into daily life. The variety of lifestyles and daily routines among residents allows the research to capture perspectives on convenience, usability, and feasibility, which are critical for designing a system that meets the needs of urban populations.

Overall, Caloocan City was chosen as the research locale because its high population, urban density, healthcare accessibility gaps, and community diversity provide a meaningful and practical setting for conducting the Concept Validation Survey. Insights obtained from this location will guide the development and refinement of the proposed IoT-based cardiac arrhythmia detection system, ensuring that it addresses real-world needs and aligns with user expectations. Urban lifestyle characteristics, which are commonly associated with increased cardiovascular risk factors such as hypertension, stress, physical inactivity, and poor dietary habits. These factors significantly contribute to the development of heart-related conditions, including cardiac arrhythmia. Arrhythmia is often considered a silent disease because symptoms may be mild, irregular, or unnoticed until serious complications arise. As a result, early detection through accessible and continuous monitoring systems is essential.

Table 1.1

Five-point Likert Scale

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

Table 1.2 demonstrates the integration of SUS with a five-point Likert scale therefore ensures that usability is measured in a structured, reliable, and academically accepted manner, strengthening the overall evaluation of the proposed system.

Population, Sample Size and Sampling Technique

This study employed purposive sampling to select respondents who met specific inclusion criteria relevant to the objectives of the research. Participants were intentionally chosen based on their medical background and experience related to cardiac health monitoring. The selected respondents included individuals diagnosed with cardiac arrhythmia and/or individuals with experience using wearable health monitoring devices capable of collecting photoplethysmography (PPG) data.

This sampling method was utilized to ensure that the data gathered and feedback obtained were directly relevant to evaluating the performance, usability, and effectiveness of the proposed Internet of Things (IoT)-based arrhythmia detection framework. Purposive sampling was deemed appropriate for this study due to the specialized nature of the target population and the need for respondents who possess specific characteristics aligned with the research objectives

Data Gathering Procedure

The initial stage of data collection starts at the user level through non-invasive optical sensing. The system utilizes a MAX30102 Photoplethysmography (PPG) Sensor, typically integrated into a wearable or wrist-worn device. The data is obtained by emitting 940 nm and red 660 nm light into the user's skin. The sensor's photodiode measures the variations in reflected light intensity, which corresponds to changes in peripheral blood volume during each cardiac cycle. These optical variations are then converted into digital signals within the sensor module. The resulting data stream consists of the raw PPG waveform, which serves as the foundational record for each pulse detected by the device. Once the raw signals are

captured, the data follows a structured path through the IoT-enabled framework to ensure it is available for both real-time viewing and long-term recording. The digital raw PPG signals are transmitted from the MAX 30102 sensor to an ESP32-C3 Microcontroller via an I2C Communication protocol, after which is initially handled by the microcontroller, the data is transmitted to the Raspberry Pi for further processing. The extracted beat-to-beat intervals and heart rate features are routed to a Raspberry Pi, which acts as the central hub for the system's machine learning operations. Simultaneously, the heart rate data, raw PPG waveforms, and detection statuses are sent to a User Interface (UI) for real-time visualization. For events where irregularities are identified, metadata including timestamps, device identifiers, and SHA-256 cryptographic hashes are stored locally in the SQLite database. When internet connectivity is available, these cached records are synced to the Hyperledger Fabric blockchain network, creating an immutable, tamper-proof record of cardiac events.

To evaluate the system, Bautista et al. (2024) recommend using the ISO/IEC 25010 framework; this approach ensures the software aligns with critical quality metrics including functionality, performance efficiency, and user interaction based on direct feedback. From the response of users, there will be feedback and evaluation based on the ISO 25010 framework. By applying this framework, the researchers ensure that the developed IoT-based cardiac monitoring system is assessed not only for technical accuracy but also for operational efficiency and user satisfaction.

Functional suitability is evaluated by determining whether the system accurately detects heart rate patterns, processes PPG signals correctly, and records irregular cardiac

events as intended. Performance efficiency is measured through response time analysis, processing latency between sensor input and user interface display, and the speed of blockchain logging during irregularity detection. Reliability testing assesses system stability during continuous monitoring sessions, including its ability to maintain consistent data transmission without signal loss or system failure.

Usability is measured through structured user feedback surveys administered to participants after interacting with the wearable device and monitoring interface. The survey instrument utilizes a Likert-scale questionnaire to evaluate interface clarity, ease of navigation, readability of displayed cardiac data, and overall user satisfaction. Security is assessed by auditing encryption mechanisms, Wired transmission safeguards, and the integrity of blockchain-recorded metadata to ensure protection against unauthorized access or tampering. Additionally, compatibility and portability are examined by testing the system across different mobile devices and operating environments, while maintainability is evaluated by analyzing the modular software structure and the ease of implementing updates or debugging procedures.

User responses are collected, tabulated, and statistically analyzed to determine mean scores for each quality characteristic under the ISO/IEC 25010 framework. The results provide quantitative and qualitative evidence of system effectiveness and guide further improvements to enhance both technical performance and user experience.

Research Instrument

To evaluate the feasibility and acceptability of the proposed IoT-based cardiac arrhythmia detection system, the researchers will utilize a Concept Validation Survey

Questionnaire as the primary research instrument. This instrument is designed to assess respondents' perceptions regarding the proposed system. The survey will determine whether the concept of the wearable device, which uses photoplethysmography (PPG) sensors and Internet of Things (IoT) technology to detect cardiac arrhythmias, is considered useful, feasible, and relevant by potential users and experts.

The use of concept validation surveys is supported by previous studies emphasizing early feedback in healthcare technology development. Mannhart et al. (2022) demonstrated that self-learned morphological features from ECG and PPG signals could correspond to actual fibrillatory waves, highlighting the importance of physiologically meaningful explanations for user trust. Likewise, Hussain et al. (2021) noted that IoT-based health monitoring systems enhance real-time data transmission and remote patient monitoring, which improves accessibility and facilitates early detection of health issues. Early concept validation, as shown in community-based studies by Lee et al. (2020), provides critical insights regarding usability, feasibility, and potential adoption, ensuring that the system aligns with user expectations and addresses practical concerns.

The survey will consist of a structured questionnaire containing statements related to the system's perceived usefulness, feasibility, expected ease of use, and innovation. Respondents will evaluate each statement using a five-point Likert scale, ranging from Strongly Disagree (1), Disagree (2), Neutral (3), Agree (4), and Strongly Agree (5). The five-point Likert scale was selected because it allows respondents to express varying degrees of agreement while maintaining clarity and simplicity. Furthermore, this format enables the researchers to perform statistical analysis such as weighted mean computation to interpret participants' evaluation of the proposed system.

This approach is consistent with best practices in healthcare technology research, where concept validation helps identify potential design challenges, usability concerns, and practical issues. By doing so, the researchers aim to ensure that the system aligns with user expectations and addresses real-world needs, increasing the likelihood of acceptance upon full implementation (Hussain et al., 2021; Mannhart et al., 2022; Lee et al., 2020).

A. Expected Ease of Use

Understanding whether participants can grasp the system concept is crucial for evaluating its potential usability. These questions measure how intuitive and logically designed the proposed IoT-based cardiac arrhythmia detection system appears before developing a prototype.

Table 1.2

Expected Ease of Use

N o .	Statement	1	2	3	4	5
1	The overall design of this system concept seems straightforward and coherent.	☐	☐	☐	☐	☐
2	The way the system is organized appears logical and understandable.	☐	☐	☐	☐	☐
3	The sequence of actions in this system concept is easy to grasp in theory.	☐	☐	☐	☐	☐
4	The features proposed in this system appear to be clearly defined.	☐	☐	☐	☐	☐

5	The concept of this system seems practical for everyday health monitoring.	☐	☐	☐	☐	☐
6	The underlying idea of the system is easy to conceptualize.	☐	☐	☐	☐	☐
7	The workflow suggested by the system concept seems sensible.	☐	☐	☐	☐	☐
8	The structure of the system appears easy to learn in principle.	☐	☐	☐	☐	☐
9	The arrangement of the system's functions appears coherent and rational.	☐	☐	☐	☐	☐
10	The concept of this system seems accessible and not overly complicated.	☐	☐	☐	☐	☐

B. Innovation

Innovation is important for demonstrating how this system differs from existing heart monitoring solutions. These questions evaluate whether participants perceive the system as novel and capable of introducing new approaches to cardiac health monitoring.

Table 1.3

Innovation

No.	Statement	1	2	3	4	5
1	The concept introduces a novel approach to heart health monitoring.	☐	☐	☐	☐	☐

2	Incorporating wearable technology with internet connectivity is an innovative idea.	☐	☐	☐	☐	☐
3	The system represents a modern perspective on health monitoring solutions.	☐	☐	☐	☐	☐
4	The concept has potential to enhance existing healthcare monitoring methods.	☐	☐	☐	☐	☐
5	The idea aligns with contemporary trends in digital health technology.	☐	☐	☐	☐	☐

C. Perceived Usefulness

Perceived usefulness reflects the potential impact of the system in addressing heart health concerns. These questions assess whether participants believe the system could support early detection, improve awareness, and provide value to users even before it is built.

Table 1.4 *Perceived Usefulness*

N o. .	Statement	1	2	3	4	5
1	This system concept could meaningfully contribute to monitoring heart health.	☐	☐	☐	☐	☐

2	The idea of a wearable device seems beneficial for individuals concerned about cardiovascular well-being.	☐	☐	☐	☐	☐	D. System
3	The system concept could support a proactive approach to general health maintenance.	☐	☐	☐	☐	☐	
4	The design concept appears to address a tangible health-related need.	☐	☐	☐	☐	☐	
5	The idea could help individuals become more aware of potential heart health issues.	☐	☐	☐	☐	☐	

Feasibility

Feasibility determines whether the concept could realistically be developed and implemented. These questions examine the practicality of building the IoT-based system using current technology, resources, and design considerations.

Table 1.5

System Feasibility

No.	Statement	1	2	3	4	5
1	The integration of sensors and internet-connected devices in this concept appears plausible.	☐	☐	☐	☐	☐
2	The proposed system design seems achievable within current technological capabilities.	☐	☐	☐	☐	☐
3	The concept appears feasible to develop and implement effectively.	☐	☐	☐	☐	☐

4	The system design seems realistic in terms of functionality and deployment.	☐	☐	☐	☐	☐
5	The proposed idea could be practically realized in a real-world context.	☐	☐	☐	☐	☐
6	The concept seems achievable without excessive complexity or resources.	☐	☐	☐	☐	☐
7	The system idea appears manageable in terms of production and maintenance.	☐	☐	☐	☐	☐
8	The technological requirements of the concept seem reasonable.	☐	☐	☐	☐	☐
9	The design concept could realistically be developed and used by intended users.	☐	☐	☐	☐	☐
10	The system concept appears implementable with existing knowledge and tools.	☐	☐	☐	☐	☐

Statistical Treatment of Data

The data collected from the respondents were systematically analyzed using appropriate statistical tools to evaluate the effectiveness, performance, and overall acceptability of the developed system. Since the study employed a survey method utilizing a five-point Likert scale, both descriptive statistics and reliability analysis were applied.

Frequency and Percentage Distribution

Frequency and percentage distribution were used to describe the profile of responses for each survey item. These statistical tools helped determine how many respondents selected a particular rating for each statement in the questionnaire.

The percentage was computed using the formula:

Where:

- f = frequency of responses
- N = total number of respondents

This method provided a clear summary of the distribution of responses across the rating scale.

Weighted Mean

The weighted mean was utilized to determine the overall evaluation of the system based on respondents' ratings using the five-point Likert scale. It provided a numerical representation of the average response for each indicator and for the system as a whole.

The formula used was:

Where:

- f = frequency of responses
- x = scale value assigned to each response
- N = total number of respondents

The computed weighted mean was interpreted using the following scale:

Table 1.6

Weighted Mean

Scale	Mean Range	Interpretation
-------	------------	----------------

5	4.21 – 5.00	Strongly Agree
4	3.41 – 4.20	Agree
3	2.61 – 3.40	Neutral
2	1.81 – 2.60	Disagree
1	1.00 - 1.80	Strongly Disagree

The computed weighted mean was interpreted using the following scale. The overall weighted mean determined the level of effectiveness, usability, and acceptability of the developed system.

Standard Deviation

Standard deviation was used to measure the variability or dispersion of the respondents' answers from the computed mean. It indicated the degree of agreement or consistency among respondents.

A **low standard deviation** signifies that responses are closely clustered around the mean, indicating consistent perceptions.

A **high standard deviation** indicates that responses are widely spread, reflecting varied opinions among respondents.

This measure complemented the weighted mean by providing insight into the reliability of the responses.

Reliability Testing (Cronbach's Alpha)

To ensure the internal consistency and reliability of the survey instrument, Cronbach's Alpha coefficient was computed. This statistical test measures how closely related a set of items are as a group, indicating the reliability of the questionnaire.

The interpretation of Cronbach's Alpha is as follows:

For this study, a reliability coefficient of **0.70 or higher** was considered acceptable, indicating that the survey instrument was sufficiently reliable for data analysis.

Table 1.7

Reliability Testing (Cronbach's Alpha)

Cronbach's Alpha	Interpretation
0.90 and above	Excellent
0.80 – 0.89	Good
0.70 – 0.79	Acceptable

0.60 – 0.69	Questionable
Below 0.60	Poor

For this study, a reliability coefficient of **0.70 or higher** was considered acceptable, indicating that the survey instrument was sufficiently reliable for data analysis.

Research Design

Descriptive Design

The primary objective of this study is to design, develop and evaluate an Internet of Things (IoT)-based framework for the early detection of Atrial Fibrillation (AF) using Photoplethysmography (PPG) and machine learning. This research implements a descriptive design to provide a comprehensive profile of the developed system describing its hardware-software architecture, its functional capabilities in non-hospital settings, and its overall quality as compared with international standards. To evaluate the system's overall quality and performance, this study utilizes the ISO/IEC 25010 Software Quality Model as the primary evaluation framework. This model is used because it provides a structured approach for assessing embedded technologies under real-world medical constraints, focusing on six key pillars. Functional Suitability evaluates the system's accuracy in identifying AF and providing diagnostic support, aiming for the high sensitivity and specificity levels documented in contemporary PPG research. Performance Efficiency measures the latency of edge-level classification to ensure clinical responsiveness without relying on continuous internet connectivity. Reliability assesses the device's maturity and its ability to maintain consistent rhythm detection in low-resource settings despite motion

artifacts or environmental noise. Security focuses on the effectiveness of blockchain protocols in protecting sensitive cardiac data and managing user anonymity. Usability examines the user-friendliness of the interface for both patients and non-specialist healthcare providers, ensuring that system outputs are interpretable and actionable. Finally, Maintainability and Portability evaluates the system's sustainability and its capacity to integrate seamlessly into primary care workflows over long-term monitoring durations.

Experimental Design

The research utilizes an experimental design to test the effectiveness and capabilities of the created IoT-based system for detecting atrial fibrillation (AF) through photoplethysmography (PPG) signals in a controlled environment. The purpose of this design is to establish the cause-and-effect relationship between the physiological signals collected from the PPG sensor and the classification output produced by the system regarding the detection of atrial fibrillation's presence or absence. The experimental method is used to confirm that the developed system operates as intended under a range of operational scenarios and signal conditions. In particular, the system will be put through pre-made test cases that mimic various cardiovascular rhythm patterns, such as regular sinus rhythm and irregular rhythm conditions linked to atrial fibrillation. After passing through the signal preprocessing and classification modules of the system, the obtained PPG signals will be examined and analyzed to determine the detection outputs.

To guarantee uniformity in signal acquisition and system performance testing, controlled experimental conditions are set up in this study. Stable sensor positioning, reduced

motion artifacts, and consistent environmental conditions throughout data collection are some of these requirements. In order to isolate system behavior and precisely evaluate the efficacy of the implemented AF detection mechanism without being influenced by outside physiological or environmental noise, the researchers maintain a controlled environment. Additionally, functionality testing using scenario-based validation procedures is incorporated into the experimental design. Every test case is intended to determine whether, given a specific input condition, the system generates the anticipated output. For example, the system should detect no AF when normal PPG signals are introduced, but irregular rhythm patterns should produce a positive AF classification. To ascertain the robustness and dependability of the system, its response to various signal quality conditions such as weak signal amplitude and the presence of noise artifacts will also be investigated. Before the developed IoT-based AF detection system is evaluated, the operational accuracy and performance efficiency will be evaluated using the results of these experimental procedures.

Table 1.8

IoT-Based AF Detection System Test Case Scenarios (Signal Classification)

Test Case Instance	Input Condition (Cause)	System Process	Expected Output
1	Normal sinus rhythm PPG signal is introduced to MAX30102 sensor	Signal preprocessing → Feature extraction → ML Classification	System classifies signal as Non-AF (0)
2	Irregular PPG waveform resembling AF rhythm is	Signal preprocessing → Feature	System classifies signal as AF Detected (1)

	introduced	extraction → ML Classification	
3	Weak amplitude PPG signal is introduced	Noise filtering applied	System still produces valid classification output
4	Noisy signal due to environmental interference	Motion artifact removal algorithm applied	AF classification remains stable
5	Continuous monitoring for 60 seconds	Edge classification executed via Raspberry Pi	Real-time AF detection is displayed
6	Interrupted signal acquisition	System re-initializes signal capture	Monitoring resumes automatically
7	Valid AF signal detected	Data transmission to database	AF event stored in time-series database

The first set of test cases evaluates the core end-to-end functionality of the AF detection system under various signal conditions to ensure reliable signal processing. The experiment involves introducing both normal sinus rhythms and irregular PPG waveforms to the MAX30102 sensor to verify that the signal preprocessing, feature extraction, and machine learning modules correctly classify the output as either Non-AF or AF Detected. Furthermore, the system is tested against weak amplitude signals and environmental noise to confirm that the noise filtering and motion artifact removal algorithms maintain stable classifications. Operational scenarios, such as continuous 60-second monitoring, automatic

resumption after signal interruption, and the successful transmission of valid AF events to the time-series database, are also assessed to guarantee real-time reliability.

Table 1.9

Hardware Integration and Sensor Acquisition Test Case

Test Case Instance	Scenario	Expected System Behavior
1	MAX30102 sensor connected to ESP32-C3	PPG waveform successfully acquired
2	Sensor placed properly on wrist	Stable BPM reading displayed
3	Sensor placement removed	System indicates signal loss
4	OLED Display connected via I ² C	BPM and AF status displayed in real-time
5	Wired connection to Raspberry Pi	PPG signal transmitted successfully
6	Battery power supplied	Device operates continuously
7	Low battery condition	System maintains safe shutdown protocol

Table 1.9 focuses on hardware integration and sensor acquisition, ensuring that the physical components of the system communicate and function correctly. The experimental scenarios verify that the sensor successfully acquires PPG waveforms when connected to the ESP32-C3 microcontroller and displays a stable beats-per-minute (BPM) reading when properly placed on the wrist, while also triggering a signal loss indication upon removal. Additional tests confirm that the I²C-connected OLED display shows the BPM and AF status in real-time, the wired connection to the Raspberry Pi successfully transmits the PPG signal

and the device operates continuously on battery power while maintaining safe shutdown protocols during low battery conditions.

Table 1.10

Machine Learning Classification Performance Test Case

Test Case Instance	Input Condition	Expected Output
1	Normal sinus rhythm PPG segment	Output: Non-AF
2	AF rhythm PPG segment (irregular inter-beat pattern)	Output: AF
3	Noisy PPG segment with motion artifact	Noise filtering applied; signal quality assessed
4	Short PPG window below minimum segment length	System buffers signal; awaits sufficient window length
5	PPG segment with high amplitude variability	Output: AF
6	Peak detection failure	System recalibrates; classification deferred
7	Valid segmented PPG window generated	Classification executed by 1D-CNN

Table 1.10 The inputs test the 1D-CNN model against preprocessed,

segmented PPG waveform windows under normal sinus rhythm conditions and under atrial fibrillation rhythm conditions. The model automatically extracts temporal and morphological features from the raw waveform through its convolutional layers, without relying on manually engineered HRV features such as SDNN or RMSSD. The tests also account for system resilience, ensuring that the Raspberry Pi recalibrates the signal upon peak detection failure and successfully executes classification once a valid segmented waveform window is available

Table 1.11

IoT Database Logging Test Case (SQLite)

Test Case Instance	Scenario	Expected Output
1	AF detected by ML model	Event stored in database
2	Normal rhythm detected	Monitoring data logged
3	Timestamped signal received	Stored in hypertable
4	Continuous monitoring active	Data updated in real-time

Test Case Instance	Scenario	Expected Output
1	AF detected by ML model	Event stored in database
2	Normal rhythm detected	Monitoring data logged
3	Timestamped signal received	Stored in hypertable
5	Network latency occurs	Edge buffering applied
6	Data retrieval request	Historical data displayed
7	Monitoring restarted	New data appended

Table 1.11 outlines the IoT database logging test cases utilizing

TimeScaleDB to evaluate the system's data management and storage capabilities over a network. These scenarios confirm that both detected AF events and normal rhythm monitoring data are accurately stored in the database, specifically utilizing hypertables to organize the timestamped signals efficiently. The experimental design also tests the system's ability to update data in real-time during continuous monitoring, apply edge buffering to prevent data loss during network latency, successfully display historical data upon a retrieval request, and append new data seamlessly when monitoring is restarted.

Table 1.12*System Usability and Interface Test Case*

Test Case Instance	User Action	Expected Output
1	User powers device ON	Monitoring begins
2	User wears device properly	Stable signal captured
3	AF detected	Alert displayed on OLED
4	Normal rhythm	Status: Normal
5	detected Dashboard	Real-time monitoring
	accessed via web	visible
6	Database queried	AF history shown

7	Monitoring stopped	System safely terminates
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Table

1.12 examines the system usability and interface, focusing on the user experience and the responsiveness of the visual outputs. The test cases validate that the system begins monitoring as soon as the user powers the device ON and captures a stable signal when the hardware is worn properly. It also verifies that the OLED screen correctly displays real-time alerts when an AF event is detected and shows a standard status indicator during regular rhythms.

Table 1.13

Blockchain-Based AF Event Logging Test Case

Test Case Instance	Input Condition (Cause)	System Process	Expected Output (Effect)
1	AF event detected by ML classifier	Event and hash stored in SQLite cache	AF event saved locally; pending blockchain sync
2	Normal rhythm detected	Monitoring data transmitted	Non-AF data recorded securely
3	Internet connection established	Pending SQLite records synced to Hyperledger Fabric	Cached AF events written to blockchain ledger
4	New AF detection occurs	Raspberry Pi initiates sync to	Data appended to distributed ledger

		blockchain network	
5	Continuous monitoring enabled	Sequential transactions generated	Monitoring history updated immutably
6	Data entry completed	Hash generated for record	Unique transaction hash created
7	Stored AF record queried	Ledger validation executed	Verified AF event retrieved

Table 1.13 details the test cases for blockchain-based AF event logging, verifying the system's capacity to securely and immutably record detection data. The primary test scenario ensures that when an AF event is detected by the machine learning classifier, the system correctly packages the AF event and its SHA-256 hash into the local SQLite database for offline storage. Similarly, the system is tested to guarantee that when a normal rhythm is detected, the monitoring data is transmitted and recorded securely to maintain a complete health history. The timestamp generated from the system clock is recorded at the moment of local storage, ensuring chronological accuracy regardless of connectivity status. Blockchain submission occurs as a deferred step when internet connectivity is available, at which point the Raspberry Pi syncs all pending cached records to the Hyperledger Fabric network and appending to the distributed ledger. The experimental design also validates that enabling continuous monitoring generates sequential transactions, updating the patient's monitoring history immutably. Upon the completion of data entry, the system is expected to generate a cryptographic hash for the record, resulting in a unique transaction hash that ensures data

integrity. Finally, the system must demonstrate that when a stored AF record is queried, it successfully executes ledger validation to retrieve the verified AF event securely.

Table 1.14
Blockchain-Based AF Event Logging Test Case

Test Case Instance	Scenario	Expected System Behavior	Expected Output (Effect)
1	AF event stored in	Record becomes immutable	AF event stored as blockchain record

	blockchain		
2	Attempt to modify stored AF data	Modification rejected	Non-AF data recorded securely
3	Duplicate AF transaction submitted	Blockchain prevents duplication	Block created with timestamped AF event
4	Ledger synchronization requested	Nodes validate transaction	Data appended to distributed ledger
5	System restart occurs	Blockchain data remains intact	Monitoring history updated immutably
6	Transaction hash verified	Data integrity confirmed	Unique transaction hash created

7	Stored monitorin g record accessed	Verified output displayed	Verified AF event retrieved
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Table 3.7 presents the blockchain-based AF event logging test cases, which are designed to verify the security, immutability, and integrity of the recorded health data. The scenarios ensure that once an AF event is stored in the blockchain, the record becomes immutable, and any attempts to modify the stored data are immediately rejected by the network. The tests also confirm that the system prevents duplicate AF transactions, successfully appends data to the distributed ledger upon node validation, maintains intact monitoring history after a system restart, creates unique transaction hashes for data integrity confirmation, and allows users to securely retrieve and view verified AF event records.

Hardware Requirements

Figure 1.1

MAX30102 PPG Sensor(Makerlab Electronics, 2026.)



The MAX30102 is an integrated optical sensor module designed for measuring photoplethysmography (PPG) signals. It incorporates infrared (IR) and red light-emitting diodes along with a photodiode to detect subtle variations in blood volume at the wrist. Communication with microcontrollers is facilitated through the I²C protocol, allowing for fast and reliable data transfer. Its small form factor, low power consumption, and continuous measurement capability make it suitable for wearable applications, particularly in monitoring heart rate and identifying irregular cardiac rhythms such as atrial fibrillation.

Figure 1.2

ESP32-C3 Microcontroller(MakerLabs Electronics, 2026.)



The ESP32-C3 is a low-power microcontroller manufactured by Espressif Systems. It transmits data using Type C USB. The microcontroller is responsible for acquiring PPG data from the MAX30102, performing preliminary signal processing, and updating the 0.96-inch OLED display with relevant metrics in real time. The ESP32-C3 supports multiple interfaces, including I²C, SPI, and UART, and its compact size, combined with low energy consumption, makes it suitable for integration into a wrist-wearable devices

Figure 1.3

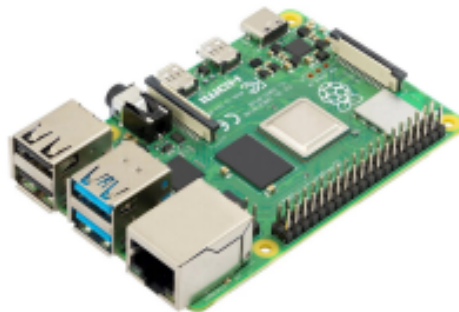
SSD1306, I²C(Makerlabs Electronics, 2026.)



The 0.96-inch OLED display is a monochrome display module with a resolution of 128×64 pixels. It is connected to the ESP32-C3 via the I²C interface, providing real-time visualization of heart rate measurements and atrial fibrillation alerts. Its lightweight construction and small dimensions make it ideal for wearable applications, allowing immediate feedback to the user while maintaining a compact form factor.

Figure 1.4

Raspberry Pi 4 Model B(Makerlabs Electronics, 2026.)



The Raspberry Pi 4 Model B is a single-board computer with a quad-core processor and 4GB or 8GB of RAM. It serves as the main on-device processing unit, running a

Python-based signal processing and inference pipeline for real-time PPG preprocessing, peak detection, and 1D-CNN arrhythmia classification. MATLAB and the PPG-Beats toolbox are used separately in the offline training environment for dataset preparation and model development. PPG data transmitted from the ESP32-C3 via Wired Transmission is received by the Raspberry Pi and processed to detect atrial fibrillation in real time. The Raspberry Pi also manages data storage and logging, providing a reliable platform for computational tasks that require higher processing capability than the microcontroller

Figure 1.5

Power Supply Li-po battery (Makerlabs Electronics, 2026)



The wearable bracelet is powered by a 3.7V rechargeable lithium-polymer (Li-Po)

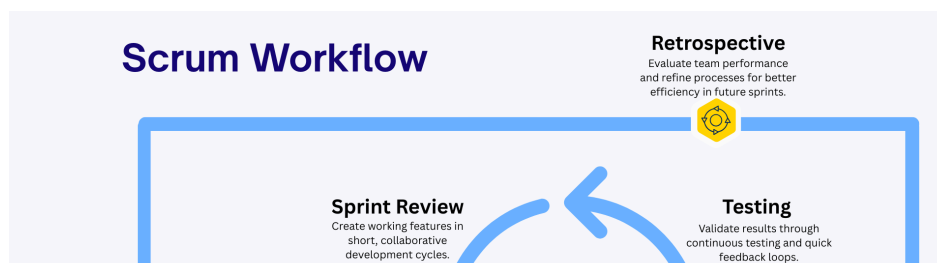
battery with a capacity of 500–1000 mAh. The battery supplies the ESP32-C3, MAX30102 sensor, and OLED display, ensuring uninterrupted operation during continuous monitoring. Its compact size and lightweight design allow for integration into a wearable form factor without compromising mobility.

Software Requirements

The researchers will adopt the Agile Scrum Methodology as its Software Development Lifecycle (SDLC) mode due to its iterative, incremental, and modular development structure. The proposed IoT-based AF Detection system consists of multiple independent modules including signal acquisition, processing, edge integration, and user interface. Agile Scrum enables systematic development and evaluation per sprint cycle. Each module undergoes planning, development, testing, and stakeholder review before proceeding to the next iteration, ensuring system reliability and functional validation.

Figure 1.6

SCRUM Methodology



The development of the proposed IoT-based PPG arrhythmia detection system follows the Agile Scrum framework, which structures the project into iterative sprint cycles consisting of planning, development, testing, review, and retrospective phases. During the Sprint Planning phase, the research team defines the specific functionalities to be developed within the sprint. This includes identifying system requirements such as real-time PPG monitoring, arrhythmia detection, and blockchain-based event logging. User stories are formulated to clarify functional objectives (e.g., enabling real-time heart monitoring and automatic detection of irregular rhythms), after which a product backlog is organized and a clear sprint goal is established, such as implementing PPG signal acquisition. The output of this phase is a defined sprint backlog with measurable objectives.

The Development or Implementation phase involves the construction of the system components aligned with the sprint goal. For the proposed system, this includes programming the ESP32 microcontroller to interface with the MAX30102 PPG sensor, implementing signal filtering and peak detection algorithms, offline training of a 1D-CNN model on the MIMIC PERform AF Dataset (Charlton et al., 2022), exporting the trained

model as a frozen inference file, and integrating it into the Raspberry Pi for on-device arrhythmia classification, generating SHA-256 cryptographic hashes for blockchain logging, and connecting processed data to a SQLite database. The primary output of this phase is a functional system module corresponding to the sprint objective.

Following implementation, the Testing phase verifies system performance and correctness. This includes validating PPG signal acquisition accuracy, confirming correct BPM computation, evaluating machine learning model performance using metrics such as a confusion matrix, measuring system latency from edge device to blockchain recording, ensuring correct blockchain logging of arrhythmia events, and conducting robustness tests under noisy signal conditions. The output of this stage consists of documented test results, including pass/fail indicators and quantitative performance metrics.

In the Sprint Review phase, the completed module is demonstrated to stakeholders, such as the research adviser or panel members. Demonstrations may include real-time visualization of PPG waveforms, arrhythmia detection outputs, blockchain hash storage verification, and presentation of system performance results. Stakeholder feedback is gathered to determine whether the increment is approved or requires refinement. Finally, during the Sprint Retrospective, the development team evaluates the sprint process by identifying strengths, challenges, and areas for improvement, such as system stability or communication latency issues. Insights from this reflection inform adjustments for the subsequent sprint cycle.

This iterative process repeats across multiple sprints, progressively enhancing the system's functionality and reliability. For instance, early sprints may focus on sensor integration and signal filtering, followed by feature extraction, machine learning, blockchain

integration, and ultimately full-system testing and validation. Through this structured yet adaptive approach, the system evolves incrementally while ensuring continuous testing, optimization, and stakeholder validation.

Figure 1.7

Python (Wikipedia, 2026)



Python is a high-level versatile programming language. It is now the modern main language to develop the Machine Learning & Signal Processing Engine for PPG Beats. It serves as the on-device backend for real-time signal processing and 1D-CNN inference, handling PPG preprocessing, arrhythmia classification, and data transmission to the dashboard and blockchain logging layer.

Figure 1.8

Google Antigravity IDE (Google, 2026)



Antigravity is an AI-Assisted IDE platform that provides a means to evolve AI-assisted coding from merely completing code to creating completely autonomous agentic workflows. This platform provides a "mission control" environment to manage multiple AI agents to plan, create, test, and deploy.

Figure 1.9

HTML5, CSS3, [Node.js](#) (Wikipedia , 2026)



The web dashboard is built using standard web technologies. HTML structures the content and CSS handle the styling and layout, ensuring a clean, responsive, and user-friendly interface. Node.js, a popular JavaScript runtime, powers the backend of the dashboard, handling server-side logic, API endpoints, and communication with the IoT devices, the Python ML service, and the database.

Figure 1.10

HyperLedgerFabric(HyperLedgerFabric.2026)



Hyperledger Fabric is a permissioned DLT or private ledger. This means that data is only accessible and modifiable by authorized users, thus maintaining the integrity and trust of the data stored on the blockchain. Each transaction is processed using predefined smart contracts that automatically enforce the rules of the blockchain. Smart contracts simplify and secure transactions by providing an organized and systematic method for processing transactions quickly and efficiently.

Figure 1.11

SQLite (Wikibooks.org.2026)



For the purpose of creating the SQLite database, the Raspberry Pi will serve as a communication link between the SQLite database and the rest of the system, allowing both clinical data and the system to be saved indefinitely with the use of the SQLite database engine. The SQLite database will provide a structured schema for storing time-stamped heart rate data, inter-beat intervals, and historical arrhythmia detection episodes along with blockchain transaction hashes. Therefore, the proposed system architecture will create an entire and verifiable history of medical records, regardless of limited resources and lack of internet connectivity.

Figure 1.12

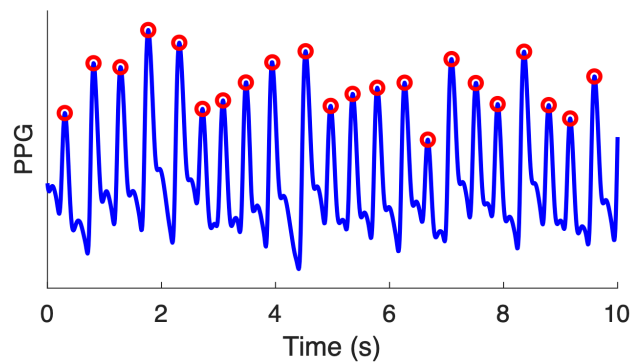
C++ (Wikipedia.2026)



The ESP32-C3 platform will serve as the hardware platform for the design of the sensor node and code will be implemented in C++ to better optimize performance of real time data acquisition and signal processing. C++ allows for very low-level hardware control, enabling high-frequency sampling loops and very efficiently optimized circular buffers to deal with the optical pulse data produced by the MAX30102 sensor with minimal latency. This will enable on-device noise filtering and preliminary feature extraction at the network's edge prior to transmitting the data outside of the network.

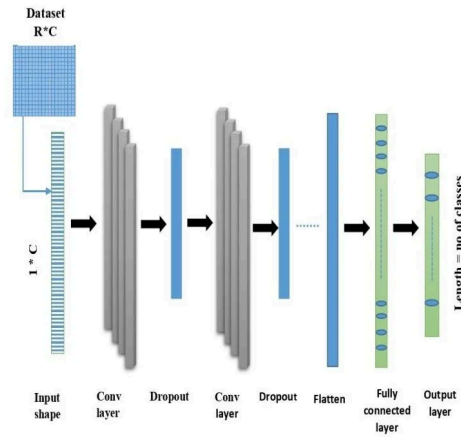
Figure 1.13

PPG Beats (PPG-beats, 2026)



PPG Beats is a software that will be used to supply researchers with algorithms for detecting heartbeats in photoplethysmography (PPG) signals, along with a means by which researchers can evaluate these algorithms performance. As such, its target audience is researched in the academic field.

Algorithms

Figure 1.14***Neural Network***

In this study, the 1D-CNN serves as the core classification engine for Atrial Fibrillation (AF) detection from photoplethysmography (PPG) signals. The model is trained offline on the MIMIC PERform AF Dataset (Charlton et al., 2022) and deployed on the Raspberry Pi for real-time edge inference. Unlike traditional machine learning approaches that require manual feature engineering (e.g., HRV statistics), the 1D-CNN automatically learns discriminative temporal and morphological features directly from segmented PPG waveforms through its convolutional layers. The model produces a binary output: 0 for Normal Sinus Rhythm (NSR) and 1 for Atrial Fibrillation (AF).

The network architecture consists of the following sequential layers. The input layer receives a preprocessed PPG signal segment of shape $(N \times 1)$. This is passed through a first Conv1D layer with 32 filters and a kernel size of 5, followed by ReLU activation and MaxPooling (pool size = 2), which extracts low-level local temporal patterns from the waveform. A second Conv1D layer with 64 filters and a kernel size of 3 then learns higher-order beat morphology and inter-beat interval irregularities, again followed by ReLU

activation and MaxPooling. The resulting feature maps are flattened into a one-dimensional vector and passed through a fully connected Dense layer of 64 neurons with ReLU activation and a Dropout rate of 0.5 to prevent overfitting. Finally, a single-neuron output layer with Sigmoid activation produces a probability value, where values ≥ 0.5 are classified as AF (label = 1) and values below 0.5 are classified as Normal Sinus Rhythm (label = 0).

The model is compiled using the Adam optimizer with binary cross-entropy as the loss function, which is standard for binary classification tasks. Training is performed offline using an 80/20 train-validation split, with early stopping applied to prevent overfitting. Once training is complete, the model weights are exported as a frozen inference file and deployed directly on the Raspberry Pi. During deployment, each incoming preprocessed PPG segment is classified in real-time in under 25 milliseconds — sufficient for continuous, low-latency arrhythmia monitoring on edge hardware without requiring an active internet connection (Cinotti et al., 2024).

Dataset

MIMIC PERform AF Dataset

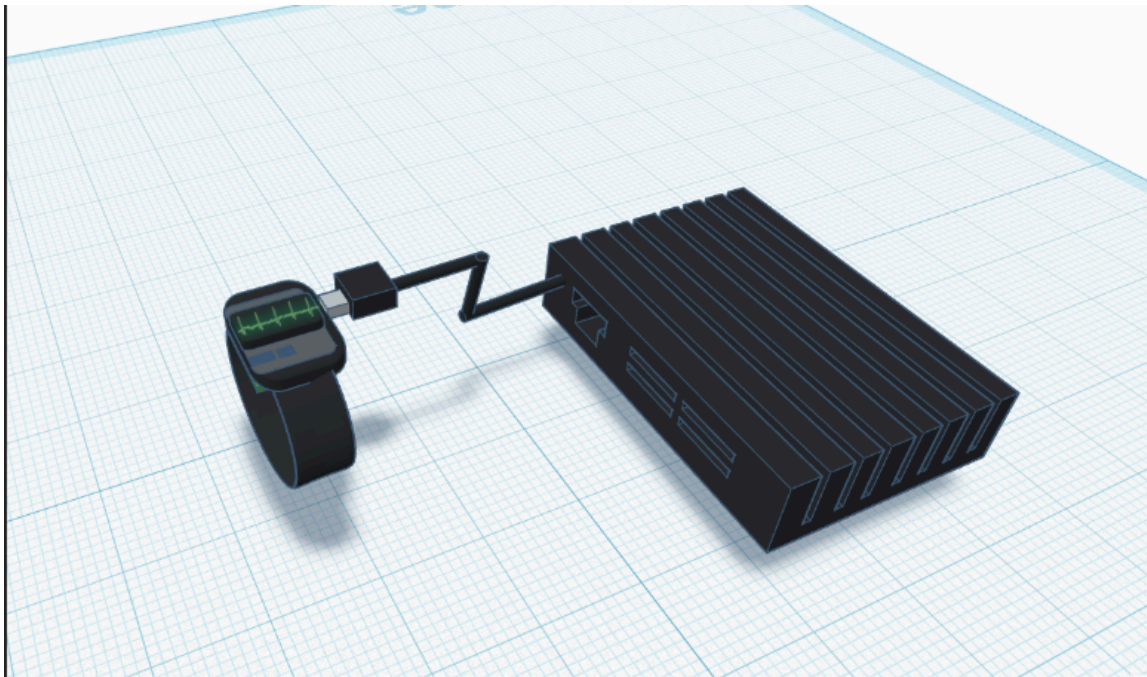
The purpose of this dataset is to benchmark the detection of Atrial Fibrillation (AF) using various physiological signals such as a Photoplethysmogram (PPG). The dataset includes real patient data and has been carefully annotated to provide synchronized data that can easily be used for research and development of wearable devices and for surveillance at the bedside.

Expected Output

The primary output of this research is a fully integrated, low-cost Internet of Things (IoT) framework designed for the real-time detection of Atrial Fibrillation (AF) through photoplethysmography (PPG) and 1D-Convolutional Neural Networks (1D-CNN).

Figure 1.16

Wearable PPG Acquisition Prototype



utilizing the MAX30102 sensor and ESP32-C3 microcontroller. It is expected to accurately capture raw PPG signals from the user's finger. The device uses red and infrared light absorption to detect blood volume changes, which are then transmitted via wire to the Raspberry Pi. The design focuses on portability and comfort, ensuring that cardiac monitoring can occur in a free-living environment without the need for bulky ECG leads.

Figure 1.17

System Dashboard

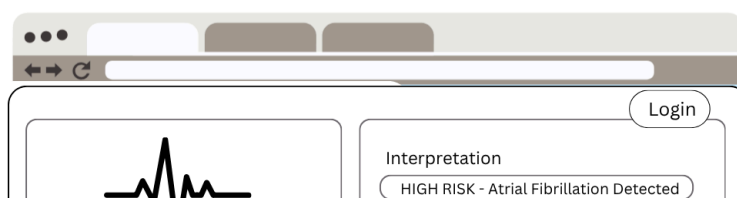
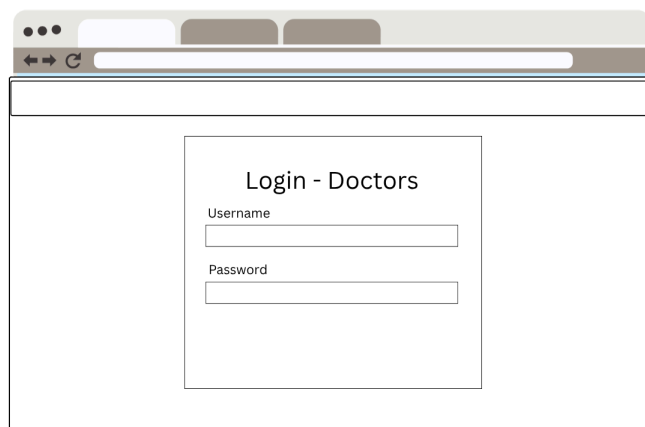


Figure 1.17 illustrates the interface providing a real-time visualization of the patient's heart rate (BPM) and PPG waveforms. It is expected to function as a clinical decision support tool, where healthcare providers can view historical data and receive instant notifications if the machine learning model detects an arrhythmia event. The UI is designed to be intuitive, allowing for patient monitoring.

Figure 1.18

Login Page



Username

Password

Figure 1.18 presents the login page designed specifically for doctors. This feature is essential in preventing unauthorized access to patient health records. It aligns with the Security characteristic of the ISO/IEC 25010 standard, which emphasizes the protection of sensitive data by ensuring that only authorized users can access confidential patient information.

System Architecture

The system outputs a seamless data pipeline. The workflow begins at the Data Acquisition stage (patient-worn device), where raw physiological signal data is collected and transmitted (as indicated by the dashed line) to the edge computing gateway (Raspberry Pi), located at the edge of the local network.

Upon receiving the transmitted signal, the edge device processes the data using a Python-based signal processing pipeline, performing noise filtering and peak detection to identify the cardiac cycle. Instead of relying solely on handcrafted HRV features for traditional machine learning models, the processed or segmented PPG signal is then fed directly into a 1D Convolutional Neural Network (1D-CNN) for rhythm classification. The 1D-CNN automatically learns relevant temporal and morphological features from the sequential signal data, eliminating the need for manual feature engineering such as HRV extraction for classification. This approach enables the model to capture complex, non-linear patterns in the physiological signals, improving the robustness and accuracy of AF detection compared to conventional methods such as Random Forest algorithms.

After machine learning processing occurs the Classification results, along with processed data, will be sent through the architecture into an SQLite database. This provides a robust offline-capable way of storing patient event log data recorded until the patient is no longer within reach of the clinician. Patient data is also transmitted via the architecture to the main dashboard, which serves as a host for all patient data. The main dashboard can only be accessed by authorized personnel through a secure log-in portal, allowing clinicians access to patient data remotely. The dashboard wireframe demonstrates the system's prioritization of clinical interpretability by providing a visual representation of a Patient's real-time PPG waveforms, providing the AF classification result and confidence output from the 1D-CNN, as well as a complete history of events recorded for that Patient, The method of architecture illustrates that beyond the local edge environment there are secure pathways established to sync locally cached SQLite data directly to the Hyperledger Fabric blockchain network once an internet connection is made, providing assured and verifiable cryptographic integrity of clinical records. This sync is initiated by the Raspberry Pi and operates independently of the dashboard.

Bibliography

- A. Bench, C., Strodthoff, N., Moulaeifard, M., Aston, P., & J. Thompson, A. (2024, December 1). Towards Trustworthy Atrial Fibrillation Classification from Wearables Data: Quantifying Model Uncertainty. 2024 Computing in Cardiology Conference. <https://doi.org/10.22489/CinC.2024.068>

- Abdelrazik, A., Eldesouky, M., Antoun, I., Lau, E. Y. M., Koya, A., Vali, Z., Suleman, S. A., Donaldson, J., & Ng, G. A. (2025). Wearable Devices for Arrhythmia Detection: Advancements and Clinical Implications. *Sensors*, 25(9), 2848. <https://doi.org/10.3390/s25092848>
- Abdulmalek, S., Nasir, A., Jabbar, W. A., Almuahaya, M. A. M., Bairagi, A. K., Khan, M. A.-M., & Kee, S.-H. (2022). IoT-Based Healthcare-Monitoring System towards Improving Quality of Life: A Review. *Healthcare*, 10(10) <https://doi.org/10.3390/healthcare10101993>.
- Adadi, A., & Berrada, M. (2018). Peeking Inside the Black-Box: A Survey on Explainable Artificial Intelligence (XAI). *IEEE Access*, 6, 52138–52160. <https://doi.org/10.1109/ACCESS.2018.2870052>
- Akram, M. S., Varma, B. S., Javed, A., Harkin, J., & Finlay, D. (2025). Toward TinyDPFL Systems for Real-Time Cardiac Healthcare: Trends, Challenges, and System-Level Perspectives on AI Algorithms, Hardware, and Edge Intelligence. Preprints. <https://doi.org/10.36227/techrxiv.174962004.42977112/v1>
- Aliamiri, A., & Shen, Y. (2018). Deep learning based atrial fibrillation detection using wearable photoplethysmography sensor. 2018 IEEE EMBS International Conference on Biomedical & Health Informatics (BHI), 442–445. <https://doi.org/10.1109/BHI.2018.8333463>
- Almarshad, M. A., Islam, M. S., Al-Ahmadi, S., & BaHammam, A. S. (2022). Diagnostic Features and Potential Applications of PPG Signal in Healthcare: A

Systematic Review. Healthcare, 10(3), 547.
<https://doi.org/10.3390/healthcare10030547>

Amini, F., Pakzad, R., Sayehmiri, K., Salari-Moghaddam, A., Asadollahi, K., & Moradpour, F. (2025). Dietary and lifestyle factors associated with cardiovascular disease risk: A structural equation modeling approach. BMC Cardiovascular Disorders, 25, 885. <https://doi.org/10.1186/s12872-025-05349-5>

Argotti, Y., Kenfaoui, Y., Baron, C., Abran, A., & Esteban, P. (2024). An Operational Quality Model of Embedded Software Aligned with ISO 25000. ACM Trans. Embed. Comput. Syst., 24(1), 9:1-9:41. <https://doi.org/10.1145/3691642>

Aschbacher, K., Yilmaz, D., Kerem, Y., Crawford, S., Benaron, D., Liu, J., Eaton, M., Tison, G. H., Olgin, J. E., Li, Y., & Marcus, G. M. (2020). Atrial fibrillation detection from raw photoplethysmography waveforms: A deep learning application. Heart Rhythm O2, 1(1), 3–9. <https://doi.org/10.1016/j.hroo.2020.02.002>

Asyrafı, H., Sunarno, Waruwu, M. M., & Wijaya, R. (2025). Analysis of Photoplethysmogram (PPG) Signal Processing Results Compared to the Graph of Electrocardiogram (ECG). JMCER, Volume 2025, 10–18.

Atrial fibrillation detection from raw photoplethysmography waveforms: A deep learning application—ScienceDirect. (2026, February 3).
https://www.sciencedirect.com/science/article/pii/S2666501820300040?utm_source=chatgpt.com

- Auton, A., Zaman, S., Padayachee, Y., Samways, J. W., Quaife, N. M., Sweeney, M., Tenorio, I., Linton, N. W. F., Cole, G. D., Peters, N. S., Mayet, J., Barton, C., & Plymen, C. (2023). Smartphone-Based Remote Monitoring for Chronic Heart Failure: Mixed Methods Analysis of User Experience From Patient and Nurse Perspectives. *JMIR Nursing*, 6(1), e44630. <https://doi.org/10.2196/44630>
- Azbeq, K., Ouchetto, O., & Jai Andaloussi, S. (2023). Access Control and Privacy-Preserving Blockchain-Based System for Diseases Management. *IEEE Transactions on Computational Social Systems*, 10(4), 1515–1527. <https://doi.org/10.1109/TCSS.2022.3186945>
- Baman, J. R., Mathew, D. T., Jiang, M., & Passman, R. S. (2022). Mobile Health for Arrhythmia Diagnosis and Management. *Journal of General Internal Medicine*, 37(1), 188–197. <https://doi.org/10.1007/s11606-021-07007-w>
- Bashar, S. K., Han, D., Hajeb-Mohammadalipour, S., Ding, E., Whitcomb, C., McManus, D. D., & Chon, K. H. (2019). Atrial Fibrillation Detection from Wrist Photoplethysmography Signals Using Smartwatches. *Scientific Reports*, 9(1), 15054. <https://doi.org/10.1038/s41598-019-49092-2>
- Bertolino, A., Faria, J. P., Lago, P., & Semini, L. (2024). Quality of Information and Communications Technology: 17th International Conference on the Quality of Information and Communications Technology, QUATIC 2024, Pisa, Italy, September 11–13, 2024, Proceedings. Springer Nature.

- Biersteker, T. E., Schali, M. J., & Treskes, R. W. (2021). Impact of Mobile Health Devices for the Detection of Atrial Fibrillation: Systematic Review. *JMIR mHealth and uHealth*, 9(4), e26161. <https://doi.org/10.2196/26161>
- Bulut, M. G., Unal, S., Hammad, M., & Pławiak, P. (2025). Deep CNN-based detection of cardiac rhythm disorders using PPG signals from wearable devices. *PLOS ONE*, 20(2), e0314154. <https://doi.org/10.1371/journal.pone.0314154>
- Cacciata, M. C., Alvarado, I., Jose, M. M., & Evangelista, L. S. (2021). Health determinants and risk factors for coronary artery disease among older Filipinos in rural communities. *European Journal of Cardiovascular Nursing*, 20(6), 565–571. <https://doi.org/10.1093/eurjcn/zvaa039>
- Cañón-Clavijo, R. E., Montenegro-Marin, C. E., Gaona-Garcia, P. A., & Ortiz-Guzmán, J. (2023). IoT Based System for Heart Monitoring and Arrhythmia Detection Using Machine Learning. *Journal of Healthcare Engineering*, 2023(1), 6401673. <https://doi.org/10.1155/2023/6401673>
- Cardiovascular diseases (CVDs). (n.d.). Retrieved March 8, 2026, from [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))
- Challenges and Limitation Analysis of an IoT-Dependent System for Deployment in Smart Healthcare Using Communication Standards Features. (n.d.). Retrieved March 8, 2026, from https://www.researchgate.net/publication/371119994_Challenges_and_Limitation

[Analysis of an IoT-Dependent System for Deployment in Smart Healthcare Using Communication Standards Features](#)

- Charlton, P. H., Kyriacou, P. A., Mant, J., Marozas, V., Chowienczyk, P., & Alastruey, J. (2022). Wearable Photoplethysmography for Cardiovascular Monitoring. *Proceedings of the IEEE*, 110(3), 355–381. <https://doi.org/10.1109/JPROC.2022.3149785>
- Chen, D., Han, D., Mercado-Díaz, L. R., Moon, J., & Chon, K. H. (2024). Smartwatch Photoplethysmogram-Based Atrial Fibrillation Detection with Premature Atrial and Ventricular Contraction Differentiation Using Densely Connected Convolutional Neural Networks. 2024 IEEE 20th International Conference on Body Sensor Networks (BSN), 1–4. <https://doi.org/10.1109/BSN63547.2024.10780734>
- Chhiba, L., Marzak, A., & Sidqui, M. (2022). Quality Attributes for Evaluating IoT Healthcare Systems. In M. Ben Ahmed, A. A. Boudhir, İ. R. Karaş, V. Jain, & S. Mellouli (Eds.), *Innovations in Smart Cities Applications Volume 5* (pp. 495–505). Springer International Publishing. https://doi.org/10.1007/978-3-030-94191-8_40
- Christopoulos, G., Attia, Z. I., Van Houten, H. K., Yao, X., Carter, R. E., Lopez-Jimenez, F., Kapa, S., Noseworthy, P. A., & Friedman, P. A. (2022). Artificial intelligence—electrocardiography to detect atrial fibrillation: Trend of probability before and after the first episode. *European Heart Journal - Digital Health*, 3(2), 228–235. <https://doi.org/10.1093/ehjdh/ztac023>

- Cinotti, E., Centracchio, J., Parlato, S., Andreozzi, E., Esposito, D., Muto, V., Bifulco, P., & Riccio, M. (2024). A Narrowband IoT Personal Sensor for Long-Term Heart Rate Monitoring and Atrial Fibrillation Detection. *Sensors*, 24(14), 4432. <https://doi.org/10.3390/s24144432>
- Cinotti, E., Gragnaniello, M., Parlato, S., Centracchio, J., Andreozzi, E., Bifulco, P., Riccio, M., & Esposito, D. (2025). An Edge AI Approach for Low-Power, Real-Time Atrial Fibrillation Detection on Wearable Devices Based on Heartbeat Intervals. *Sensors*, 25(23). <https://doi.org/10.3390/s25237244>
- Cook, D. A., Enders, F., Caraballo, P. J., Nishimura, R. A., & Lloyd, F. J. (2015). An Automated Clinical Alert System for Newly-Diagnosed Atrial Fibrillation. *PLOS ONE*, 10(4), e0122153. <https://doi.org/10.1371/journal.pone.0122153>
- da Costa, C. A., Zeiser, F. A., da Rosa Righi, R., Antunes, R. S., Alegretti, A. P., Bertoni, A. P., de Oliveira Ramos, G., de Mello, B. H., Vanin, F., Bertoletti, O. A., & Rigo, S. J. (2024). Internet of Things and Machine Learning for Smart Healthcare. In S. Namasudra (Ed.), *IoT and ML for Information Management: A Smart Healthcare Perspective* (pp. 95–133). Springer Nature. https://doi.org/10.1007/978-981-97-5624-7_3
- Dept. of Biomedical Engineering, Udaya School of Engineering, Vellamodi, Tamil Nadu, India, R, S., & Kumar, R. M. (2023). ATRIAL FIBRILLATION RISK PREDICTION FROM ELECTROCARDIOGRAM AND RELATED HEALTH DATA WITH DEEP NEURAL NETWORK. *INTERANTIONAL JOURNAL OF*

Diagnostic accuracy of ECG smart chest patches versus PPG smartwatches for atrial fibrillation detection: A systematic review and meta-analysis | BMC Cardiovascular Disorders | Springer Nature Link. (n.d.). Retrieved February 3, 2026, from <https://link.springer.com/article/10.1186/s12872-025-04582-2>

Ding, C., Guo, Z., Rudin, C., Xiao, R., Shah, A., Do, D. H., Lee, R. J., Clifford, G., Nahab, F. B., & Hu, X. (2024). Learning From Alarms: A Robust Learning Approach for Accurate Photoplethysmography-Based Atrial Fibrillation Detection Using Eight Million Samples Labeled With Imprecise Arrhythmia Alarms. *IEEE Journal of Biomedical and Health Informatics*, 28(5), 2650–2661. <https://doi.org/10.1109/JBHI.2024.3360952>

Ding, C., Xiao, R., Wang, W., Holdsworth, E., & Hu, X. (2024). Photoplethysmography based atrial fibrillation detection: A continually growing field. *Physiological Measurement*, 45(4), 04TR01. <https://doi.org/10.1088/1361-6579/ad37ee>

Ebrahimi, Z., & Gosselin, B. (2023). Ultralow-Power Photoplethysmography (PPG) Sensors: A Methodological Review. *IEEE Sensors Journal*, 23(15), 16467–16480. <https://doi.org/10.1109/JSEN.2023.3284818>

Eerikainen, L. M., Bonomi, A. G., Schipper, F., Dekker, L. R. C., de Morree, H. M., Vullings, R., & Aarts, R. M. (2020). Detecting Atrial Fibrillation and Atrial Flutter

- in Daily Life Using Photoplethysmography Data. *IEEE Journal of Biomedical and Health Informatics*, 24(6), 1610–1618. <https://doi.org/10.1109/JBHI.2019.2950574>
- Egala, B. S., Pradhan, A. K., Badarla, V., & Mohanty, S. P. (2021). Fortified-Chain: A Blockchain-Based Framework for Security and Privacy-Assured Internet of Medical Things With Effective Access Control. *IEEE Internet of Things Journal*, 8(14), 11717–11731. <https://doi.org/10.1109/JIOT.2021.3058946>
- Fadreguilan, E. (2023). The Management of Atrial Fibrillation in the Philippines: A State-of-the-Art Review. *Philippine Journal of Cardiology*, 51(2), 80–84. <https://doi.org/10.69944/pjc.4362c3ece7>
- Fernández-Ruiz, I. (2019). AI can help to diagnose AF during sinus rhythm. *Nature Reviews Cardiology*, 16(10), 578–578. <https://doi.org/10.1038/s41569-019-0256-4>
- Filgueiras, T. P., Bertemes-Filho, P., & Noveletto, F. (2025). Evaluating the Accuracy of Low-Cost Wearable Sensors for Healthcare Monitoring. *Micromachines*, 16(7). <https://doi.org/10.3390/mi16070791>
- Frontiers | Burden of atrial fibrillation and its attributable risk factors from 1990 to 2019: An analysis of the Global Burden of Disease study 2019. (n.d.). Retrieved March 8, 2026, from <https://www.frontiersin.org/journals/cardiovascular-medicine/articles/10.3389/fcvm.2022.997698/full>
- Fujiwara, K., Kawaji, T., Saeda, S., Yamakawa, T., Kato, M., Aizawa, T., Yokomatsu, T., & Miki, S. (2025). IoT-Based Atrial Fibrillation Screening System, Development,

and Prospective Study. IEEE Internet of Things Journal, 12(11), 17661–17673.
<https://doi.org/10.1109/JIOT.2025.3537636>

Girón, N. A., Millán, C. A., & López, D. M. (2019). Systematic Review on Features Extracted from PPG Signals for the Detection of Atrial Fibrillation. Studies in Health Technology and Informatics, 261, 266–273.

Gopinathannair, R., Merchant, F. M., Lakkireddy, D. R., Etheridge, S. P., Feigofsky, S., Han, J. K., Kabra, R., Natale, A., Poe, S., Saha, S. A., & Russo, A. M. (2020). COVID-19 and cardiac arrhythmias: A global perspective on arrhythmia characteristics and management strategies. Journal of Interventional Cardiac Electrophysiology: An International Journal of Arrhythmias and Pacing, 59(2), 329–336. <https://doi.org/10.1007/s10840-020-00789-9>

Gruwez, H., Ezzat, D., Van Puyvelde, T., Dhont, S., Meekers, E., Bruckers, L., Wouters, F., Kellens, M., Van Herendael, H., Rivero-Ayerza, M., Nuyens, D., Haemers, P., & Pison, L. (2024). Real-world validation of smartphone-based photoplethysmography for rate and rhythm monitoring in atrial fibrillation. EP Europace, 26(4), euae065. <https://doi.org/10.1093/europace/euae065>

Guarducci, S., Jayousi, S., Caputo, S., & Mucchi, L. (2025). Key Fundamentals and Examples of Sensors for Human Health: Wearable, Non-Continuous, and Non-Contact Monitoring Devices. Sensors, 25(2).
<https://doi.org/10.3390/s25020556>

- Guo, Y., Wang, H., Zhang, H., Liu, T., Li, L., Liu, L., Chen, M., Chen, Y., & Lip, G. Y. H. (2021). Photoplethysmography-Based Machine Learning Approaches for Atrial Fibrillation Prediction. *JACC: Asia*, 1(3), 399–408.
<https://doi.org/10.1016/j.jacasi.2021.09.004>
- Hammer, A., Malberg, H., & Schmidt, M. (2024, December 1). Morphology Features Self-Learned by Explainable Deep Learning for Atrial Fibrillation Detection Correspond to Fibrillatory Waves. 2024 Computing in Cardiology Conference.
<https://doi.org/10.22489/CinC.2024.305>
- Han, D., Moon, J., Díaz, L. R. M., Chen, D., Williams, D., Ding, E. Y., Tran, K.-V., McManus, D. D., & Chon, K. H. (2024). Multiclass Arrhythmia Classification using Smartwatch Photoplethysmography Signals Collected in Real-life Settings.
<https://doi.org/10.48550/ARXIV.2409.06147>
- Han, D., Moon, J., Díaz, L. R. M., Chen, D., Williams, D., Mohagheghian, F., Ghetia, O., Peitzsch, A. G., Kong, Y., Nishita, N., Ghutadaria, O., Orwig, T. A., Otabil, E. M., Noorishirazi, K., Hamel, A., Dickson, E. L., DiMezza, D., Lessard, D., Wang, Z., ... Chon, K. H. (2025). Multiclass Arrhythmia Classification using Multimodal Smartwatch Photoplethysmography Signals Collected in Real-life Settings. *IEEE Transactions on Biomedical Engineering*, 1–14.
<https://doi.org/10.1109/TBME.2025.3613471>
- Hayiroğlu, M. İ., & Altay, S. (2023). The Role of Artificial Intelligence in Coronary Artery Disease and Atrial Fibrillation. *Balkan Medical Journal*, 40(3), 151–152.
<https://doi.org/10.4274/balkanmedj.galenos.2023.06042023>

- Hermans, A. N. L., Gawalko, M., Dohmen, L., Van Der Velden, R. M. J., Betz, K., Duncker, D., Verhaert, D. V. M., Heidebuchel, H., Svennberg, E., Neubeck, L., Eckstein, J., Lane, D. A., Lip, G. Y. H., Crijns, H. J. G. M., Sanders, P., Hendriks, J. M., Pluymaekers, N. A. H. A., & Linz, D. (2022). Mobile health solutions for atrial fibrillation detection and management: A systematic review. *Clinical Research in Cardiology*, 111(5), 479–491. <https://doi.org/10.1007/s00392-021-01941-9>
- Hermans, A. N. L., Gawalko, M., Dohmen, L., Van Der Velden, R. M. J., Betz, K., Verhaert, D. V. M., Pluymaekers, N. A. H. A., Hendriks, J. M., & Linz, D. (2022). A systematic review of mobile health opportunities for atrial fibrillation detection and management. *European Journal of Preventive Cardiology*, 29(5), e205–e208. <https://doi.org/10.1093/euripc/zwab158>
- Hiraoka, D., Inui, T., Kawakami, E., Oya, M., Tsuji, A., Honma, K., Kawasaki, Y., Ozawa, Y., Shiko, Y., Ueda, H., Kohno, H., Matsuura, K., Watanabe, M., Yakita, Y., & Matsumiya, G. (2022). Diagnosis of Atrial Fibrillation Using Machine Learning With Wearable Devices After Cardiac Surgery: Algorithm Development Study. *JMIR Formative Research*, 6(8), e35396. <https://doi.org/10.2196/35396>
- Howlader, R., Jurgec, M., Schimanko, A., Dohnal, C., Busch, A., Baumgartner, M., Wiesmüller, F., Lauschensky, A., Hayn, D., & Schreier, G. (2025). A Neural Network for Atrial Fibrillation Detection via PPG. In M. Baumgartner, D. Hayn, B. Pfeifer, & G. Schreier (Eds.), *Studies in Health Technology and Informatics*. IOS Press. <https://doi.org/10.3233/SHTI250201>

- Ibrahim Obaid, O., & Salman, S. (2023). Security and Privacy in IoT-based Healthcare Systems: A Review. *Mesopotamian Journal of Computer Science*, 2022. <https://doi.org/10.58496/MJCSC/2022/007>
- Iqbal, M. S., Belal Bin Heyat, M., Parveen, S., Ammar Bin Hayat, M., Roshanzamir, M., Alizadehsani, R., Akhtar, F., Sayeed, E., Hussain, S., Hussein, H. S., & Sawan, M. (2024). Progress and trends in neurological disorders research based on deep learning. *Computerized Medical Imaging and Graphics*, 116, 102400. <https://doi.org/10.1016/j.compmedimag.2024.102400>
- Iqhrammullah, M., Abdullah, A., Hermansyah, Ichwansyah, F., Rani, H. A., Alina, M., Simanjuntak, A. M. T., Rampengan, D. D. C. H., Al-Gunaid, S. T., Gusti, N., Damarkusuma, A., & Wikurendra, E. A. (2025). Accuracy and interpretability of smartwatch electrocardiogram for early detection of atrial fibrillation: A systematic review and meta-analysis. *Journal of Arrhythmia*, 41(3), e70087. <https://doi.org/10.1002/joa3.70087>
- Jahmunah, V., Ng, E. Y. K., Tan, R.-S., Oh, S. L., & Acharya, U. R. (2022). Explainable detection of myocardial infarction using deep learning models with Grad-CAM technique on ECG signals. *Computers in Biology and Medicine*, 146, 105550. <https://doi.org/10.1016/j.compbiomed.2022.105550>
- Jan 10, L. R. & 2024. (n.d.). What Is Cardiovascular Disease? [Www.Heart.Org](https://www.heart.org/en/health-topics/consumer-healthcare/what-is-cardiovascular-disease). Retrieved March 8, 2026, from <https://www.heart.org/en/health-topics/consumer-healthcare/what-is-cardiovascular-disease>

Jeanningros, "Loïc, Braun, F., Van Zaen, J., Le Bloa, M., Porretta, A., Teres, C., Herrera, C., Domenichini, G., Carroz, P., Graf, D., Pascale, P., Vesin, J.-M., Thiran, J.-P., Pruvot, E., & Lemay", M. (2022, December 31). Pulse Wave Analysis of Photoplethysmography Signals to Enhance Classification of Cardiac Arrhythmias. 2022 Computing in Cardiology Conference. <https://doi.org/10.22489/CinC.2022.023>

Jia, B., Chen, J., Luan, Y., Wang, H., Wei, Y., & Hu, Y. (2024). Artificial intelligence and atrial fibrillation: A bibliometric analysis from 2013 to 2023. *Heliyon*, 10(15), e35067. <https://doi.org/10.1016/j.heliyon.2024.e35067>

Jia, Y., Jiang, H., Yang, P., & He, X. (2020, December 30). An End-to-end Deep Learning Scheme for Atrial Fibrillation Detection. 2020 Computing in Cardiology Conference. <https://doi.org/10.22489/CinC.2020.106>

JMIR Cardio—Photoplethysmography-Based Machine Learning Approaches for Atrial Fibrillation Burden: Algorithm Development and Validation. (2026, February 3). https://cardio.jmir.org/2025/1/e78075?utm_source=chatgpt.com

Karanikola, A.-E., Tsiachris, D., Argyriou, N., Botis, M., Pamporis, K., Xydis, P., Fragoulis, C., Kordalis, A., & Tsioufis, K. (2025). Complex interrelationship and therapeutic advances in diabetic patients with atrial fibrillation. *World Journal of Cardiology*, 17(12), 114575. <https://doi.org/10.4330/wjc.v17.i12.114575>

- Kingma, J., Simard, C., & Drolet, B. (2023). Overview of Cardiac Arrhythmias and Treatment Strategies. *Pharmaceuticals*, 16(6), 844. <https://doi.org/10.3390/ph16060844>
- Klima, M., Bures, M., Frajtak, K., Rechtberger, V., Trnka, M., Bellekens, X., Cerny, T., & Ahmed, B. S. (2022). Selected Code-Quality Characteristics and Metrics for Internet of Things Systems. *IEEE Access*, 10, 46144–46161. <https://doi.org/10.1109/ACCESS.2022.3170475>
- Kozłowski, T., Noran, O., & Trevathan, J. (2023). Designing an Evaluation Framework for IoT Environmental Monitoring Systems. *Procedia Computer Science, CENTERIS – International Conference on ENTERprise Information Systems / ProjMAN – International Conference on Project MANagement / HCist – International Conference on Health and Social Care Information Systems and Technologies 2022*, 219, 220–227. <https://doi.org/10.1016/j.procs.2023.01.284>
- Kumar, D., Peimankar, A., Sharma, K., Domínguez, H., Puthusserypady, S., & Bardram, J. E. (2022). Deepaware: A hybrid deep learning and context-aware heuristics-based model for atrial fibrillation detection. *Computer Methods and Programs in Biomedicine*, 221, 106899. <https://doi.org/10.1016/j.cmpb.2022.106899>
- Kwon, S., Hong, J., Choi, E.-K., Lee, B., Baik, C., Lee, E., Jeong, E.-R., Koo, B.-K., Oh, S., & Yi, Y. (2020). Detection of Atrial Fibrillation Using a Ring-Type Wearable Device (CardioTracker) and Deep Learning Analysis of Photoplethysmography Signals: Prospective Observational Proof-of-Concept Study. *Journal of Medical Internet Research*, 22(5), e16443. <https://doi.org/10.2196/16443>

- Lei, N., Kareem, M., Moon, S. K., Ciaccio, E. J., Acharya, U. R., & Faust, O. (2021). Hybrid Decision Support to Monitor Atrial Fibrillation for Stroke Prevention. *International Journal of Environmental Research and Public Health*, 18(2), 813. <https://doi.org/10.3390/ijerph18020813>
- Liao, M.-T., Yu, C.-C., Lin, L.-Y., Pan, K.-H., Tsai, T.-H., Wu, Y.-C., & Liu, Y.-B. (2022). Impact of recording length and other arrhythmias on atrial fibrillation detection from wrist photoplethysmogram using smartwatches. *Scientific Reports*, 12(1), 5364. <https://doi.org/10.1038/s41598-022-09181-1>
- Lingawi, S. (2025). Improving wearable out-of-hospital pulse oximetry estimation with novel real-time analysis of photoplethysmography [University of British Columbia]. <https://doi.org/10.14288/1.0451182>
- Lippi, G., Sanchis-Gomar, F., & Cervellin, G. (2021). Global epidemiology of atrial fibrillation: An increasing epidemic and public health challenge. *International Journal of Stroke*, 16(2), 217–221. <https://doi.org/10.1177/1747493019897870>
- Liu, B., Zhang, Z., Di, X., Wang, X., Xie, L., Xie, W., & Zhang, J. (2021). The Assessment of Autonomic Nervous System Activity Based on Photoplethysmography in Healthy Young Men. *Frontiers in Physiology*, 12, 733264. <https://doi.org/10.3389/fphys.2021.733264>
- Liu, Z., Zhou, B., Jiang, Z., Chen, X., Li, Y., Tang, M., & Miao, F. (2022). Multiclass Arrhythmia Detection and Classification From Photoplethysmography Signals

- Using a Deep Convolutional Neural Network. *Journal of the American Heart Association*, 11(7), e023555. <https://doi.org/10.1161/JAHA.121.023555>
- Liu, Z., Zhu, T., Lu, L., Zhang, Y., & Clifton, D. A. (2024). Intelligent Electrocardiogram Acquisition Via Ubiquitous Photoplethysmography Monitoring. *IEEE Journal of Biomedical and Health Informatics*, 28(3), 1321–1330. <https://doi.org/10.1109/JBHI.2023.3344187>
- Lown, M., Brown, M., Brown, C., Yue, A. M., Shah, B. N., Corbett, S. J., Lewith, G., Stuart, B., Moore, M., & Little, P. (2020). Machine learning detection of Atrial Fibrillation using wearable technology. *PLOS ONE*, 15(1), e0227401. <https://doi.org/10.1371/journal.pone.0227401>
- Lubitz, S. A., Faranesh, A. Z., Selvaggi, C., Atlas, S. J., McManus, D. D., Singer, D. E., Pagoto, S., McConnell, M. V., Pantelopoulos, A., & Foulkes, A. S. (2022). Detection of Atrial Fibrillation in a Large Population Using Wearable Devices: The Fitbit Heart Study. *Circulation*, 146(19), 1415–1424. <https://doi.org/10.1161/CIRCULATIONAHA.122.060291>
- Luo, J., Taleban, A., Noffke, P., & Sparapani, R. (2026). Multi-Level Explainable AI for ECG-Based Atrial Fibrillation Detection: Exploring LIME, SHAP, and Grad-CAM for Clinical Interpretation. *Proceedings of the 2025 9th International Conference on Medical and Health Informatics, ICMHI '25*, 52–59. <https://doi.org/10.1145/3761712.3761721>

- Machine Learning for Cardiovascular Healthcare: Opportunities, Challenges, and the Path Forward. (2024). *Journal of Angiotherapy*, 8(12), 1–11. <https://doi.org/10.25163/angiotherapy.81210205>
- Mahim, S. M., Emamul Hossen, M., Al Hasan, S., Islam, M. K., Iqbal, Z., Alibakhshikenari, M., Collotta, M., & Miah, M. S. (2024). TransMixer-AF: Advanced Real-Time Detection of Atrial Fibrillation Utilizing Single-Lead Electrocardiogram Signals. *IEEE Access*, 12, 143149–143162. <https://doi.org/10.1109/ACCESS.2024.3467181>
- Mäkynen, M., Ng, G. A., Li, X., Schlindwein, F. S., & Pearce, T. C. (2024). Compressed Deep Learning Models for Wearable Atrial Fibrillation Detection through Attention. *Sensors*, 24(15), 4787. <https://doi.org/10.3390/s24154787>
- Mannhart, D., Lefebvre, B., Gardella, C., Henry, C., Serban, T., Knecht, S., Kuehne, M., Sticherling, C., & Badertscher, P. (n.d.). *E-Cardiology / Digital Health, Public Health, Health Economics, Research Methodology – e-Cardiology / Digital Health, Artificial Intelligence (Machine Learning, Deep Learning)*.
- Martin. (n.d.). Health. United Nations Sustainable Development. Retrieved March 8, 2026, from <https://www.un.org/sustainabledevelopment/health/>
- Mehak, S., Ramos, I. F., Sagar, K., Ramasubramanian, A., Kelleher, J. D., Guilfoyle, M., Gianini, G., Damiani, E., & Leva, M. C. (2024). A roadmap for improving data quality through standards for collaborative intelligence in human-robot applications. *Frontiers in Robotics and AI*, 11. <https://doi.org/10.3389/frobt.2024.1434351>

- Millán, C. A., Girón, N. A., & Lopez, D. M. (2020). Analysis of Relevant Features from Photoplethysmographic Signals for Atrial Fibrillation Classification. *International Journal of Environmental Research and Public Health*, 17(2), 498. <https://doi.org/10.3390/ijerph17020498>
- Moorthy, P., Weinert, L., Schüttler, C., Svensson, L., Sedlmayr, B., Müller, J., & Nagel, T. (2024). Attributes, Methods, and Frameworks Used to Evaluate Wearables and Their Companion mHealth Apps: Scoping Review. *JMIR mHealth and uHealth*, 12(1), e52179. <https://doi.org/10.2196/52179>
- Nguyen, D. C., Ding, M., Pathirana, P. N., & Seneviratne, A. (2021). Blockchain and AI-Based Solutions to Combat Coronavirus (COVID-19)-Like Epidemics: A Survey. *Ieee Access*, 9, 95730. <https://doi.org/10.1109/ACCESS.2021.3093633>
- Norbert, N., & András, K. (n.d.). A SZOFTVERMINŐSÉG GEL KAPCSOLATOS ISO SZABVÁNYOK ÁTTEKINTÉSE. 3(2).
- Pedrosa-Rodriguez, A., Camara, C., & Peris-Lopez, P. (2024). Leveraging IoT Devices for Atrial Fibrillation Detection: A Comprehensive Study of AI Techniques. *Applied Sciences*, 14(19), 8945. <https://doi.org/10.3390/app14198945>
- Pereira, T., Tran, N., Gadhomi, K., Pelter, M. M., Do, D. H., Lee, R. J., Colorado, R., Meisel, K., & Hu, X. (2020). Photoplethysmography based atrial fibrillation detection: A review. *Npj Digital Medicine*, 3(1), 3. <https://doi.org/10.1038/s41746-019-0207-9>

- Peris-Lopez, P., Fuster-Barceló, C., Camara, C., & Martin, H. (2023). Unveiling Hidden Patterns: Harnessing the Power of Short PPG-Traces for Atrial Fibrillation Detection. 2023 IEEE 19th International Conference on Body Sensor Networks (BSN), 1–4. <https://doi.org/10.1109/BSN58485.2023.10331002>
- Phukan, N., Manikandan, M. S., & Pachori, R. B. (2025). Resource-Efficient CNN Based Atrial Fibrillation Detection Using P-Wave and RR Interval Features for Edge-AI Cardiac Health Monitoring Devices. IEEE Transactions on Artificial Intelligence, 6(10), 2754–2765. <https://doi.org/10.1109/TAI.2025.3556995>
- Pluymaekers, N. A. H. A., Hermans, A. N. L., Van Der Velden, R. M. J., Gawalko, M., Den Uijl, D. W., Buskes, S., Vernooy, K., Crijns, H. J. G. M., Hendriks, J. M., & Linz, D. (2021). Implementation of an on-demand app-based heart rate and rhythm monitoring infrastructure for the management of atrial fibrillation through teleconsultation: TeleCheck-AF. EP Europace, 23(3), 345–352. <https://doi.org/10.1093/europace/euaa201>
- Proesmans, T., Mortelmans, C., Van Haelst, R., Verbrugge, F., Vandervoort, P., & Vaes, B. (2019). Mobile Phone–Based Use of the Photoplethysmography Technique to Detect Atrial Fibrillation in Primary Care: Diagnostic Accuracy Study of the FibriCheck App. JMIR mHealth and uHealth, 7(3), e12284. <https://doi.org/10.2196/12284>
- Qananwah, Q., Ababneh, M., & Dagamseh, A. (2024). Cardiac arrhythmias classification using photoplethysmography database. Scientific Reports, 14, 3355. <https://doi.org/10.1038/s41598-024-53142-9>

Quality characteristics in IoT systems: Learnings from an industry multi case study |

Discover Internet of Things | Springer Nature Link. (n.d.). Retrieved February 3,

2026, from <https://link.springer.com/article/10.1007/s43926-025-00094-9>

Ricci, F., Mattei, E., Calcagnini, G., & Censi, F. (2025). Home detection of atrial fibrillation using cardiac activity analysis: Technologies available to the patient.

Expert Review of Medical Devices, 22(7), 711–724.

<https://doi.org/10.1080/17434440.2025.2510537>

Roth, G. A., Mensah, G. A., Johnson, C. O., Addolorato, G., Ammirati, E., Baddour, L.

M., Barengo, N. C., Beaton, A. Z., Benjamin, E. J., Benziger, C. P., Bonny, A.,

Brauer, M., Brodmann, M., Cahill, T. J., Carapetis, J., Catapano, A. L., Chugh, S.

S., Cooper, L. T., Coresh, J., ... GBD-NHLBI-JACC Global Burden of

Cardiovascular Diseases Writing Group. (2020). Global Burden of Cardiovascular

Diseases and Risk Factors, 1990-2019: Update From the GBD 2019 Study. Journal

of the American College of Cardiology, 76(25), 2982–3021.

<https://doi.org/10.1016/j.jacc.2020.11.010>

RS Lyimo,. (2025). A Portable and Affordable Photoplethysmography (PPG)-Based

Health Monitoring Solution for Low-Resource Settings. Journal of Research

Innovation and Implications in Education. <https://doi.org/10.59765/vhry46r>

Saarinen, H. J., Joutsen, A., Korpi, K., Halkola, T., Nurmi, M., Hernesniemi, J., &

Vehkaoja, A. (2023). Wrist-worn device combining PPG and ECG can be reliably

used for atrial fibrillation detection in an outpatient setting. Frontiers in

Cardiovascular Medicine, 10, 1100127. <https://doi.org/10.3389/fcvm.2023.1100127>

- Sabbadini, R., Riccio, M., Maresca, L., Irace, A., & Breglio, G. (2022). Atrial Fibrillation Detection by Means of Edge Computing on Wearable Device: A Feasibility Assessment. 2022 IEEE International Symposium on Medical Measurements and Applications (MeMeA), 1–6.
<https://doi.org/10.1109/MeMeA54994.2022.9856438>
- Salvi, M., Acharya, M. R., Seoni, S., Faust, O., Tan, R., Barua, P. D., García, S., Molinari, F., & Acharya, U. R. (2024). Artificial intelligence for atrial fibrillation detection, prediction, and treatment: A systematic review of the last decade (2013–2023). WIREs Data Mining and Knowledge Discovery, 14(3), e1530.
<https://doi.org/10.1002/widm.1530>
- Santala, O. E., Lipponen, J. A., Jäntti, H., Rissanen, T. T., Tarvainen, M. P., Laitinen, T. P., Laitinen, T. M., Castrén, M., Väliäho, E.-S., Rantula, O. A., Naukkarinen, N. S., Hartikainen, J. E. K., Halonen, J., & Martikainen, T. J. (2022). Continuous mHealth Patch Monitoring for the Algorithm-Based Detection of Atrial Fibrillation: Feasibility and Diagnostic Accuracy Study. JMIR Cardio, 6(1), e31230.
<https://doi.org/10.2196/31230>
- Sehrawat, O., Kashou, A. H., & Noseworthy, P. A. (2022). Artificial intelligence and atrial fibrillation. Journal of Cardiovascular Electrophysiology, 33(8), 1932–1943.
<https://doi.org/10.1111/jce.15440>
- Shao, M., Zhou, Z., Bin, G., Bai, Y., & Wu, S. (2020). A Wearable Electrocardiogram Telemonitoring System for Atrial Fibrillation Detection. Sensors, 20(3), 606.
<https://doi.org/10.3390/s20030606>

- SiamAF: Learning Shared Information from ECG and PPG Signals for Robust Atrial Fibrillation Detection. (2026, February 3).
https://arxiv.org/html/2310.09203v2?utm_source=chatgpt.com
- Sideshwar, J. B., Sachin Krishan, T., Nagarajan, V., S, S., & Vijayaraghavan, V. (2021). End-to-End Optimized Arrhythmia Detection Pipeline using Machine Learning for Ultra-Edge Devices. 2021 20th IEEE International Conference on Machine Learning and Applications (ICMLA), 1501–1506.
<https://doi.org/10.1109/ICMLA52953.2021.00242>
- Soudani, A. (2023). Wearable Smart Solution for Accurate Machine Learning-Based AF Detection Using ECG Signals. Engineering.
<https://doi.org/10.20944/preprints202312.1534.v1>
- Souza-Pereira, L., Ouhbi, S., & Pombo, N. (2021). A process model for quality in use evaluation of clinical decision support systems. Journal of Biomedical Informatics, 123, 103917. <https://doi.org/10.1016/j.jbi.2021.103917>
- Steinhubl, S. R., Waalen, J., Edwards, A. M., Ariniello, L. M., Mehta, R. R., Ebner, G. S., Carter, C., Baca-Motes, K., Felicione, E., Sarich, T., & Topol, E. J. (2018). Effect of a Home-Based Wearable Continuous ECG Monitoring Patch on Detection of Undiagnosed Atrial Fibrillation: The mSToPS Randomized Clinical Trial. JAMA, 320(2), 146. <https://doi.org/10.1001/jama.2018.8102>
- Sumathi, M., Vijayaraj, N., Raja, S. P. R., & Rajkamal, M. (2022). Internet of Thing Based Confidential Healthcare Data Storage, Access Control and Monitoring Using

- Blockchain Technique. Computing and Informatics, 41(5), 1207–1239.
https://doi.org/10.31577/cai_2022_5_1207
- SYNTHESIS. (2026, February 10). Google Docs.
https://docs.google.com/document/u/1/d/163ySdpWDGil6VC7FIasDaRwM5S_aklbYdDY8NVHaGCO/edit?usp=sharing&usp=embed_facebook
- Talukdar, D., de Deus, L. F., & Sehgal, N. (n.d.). Evaluation of Atrial Fibrillation Detection in short-term Photoplethysmography (PPG) signals using artificial intelligence.
- Tooley, J. E., & Perez, M. V. (2022). Role of digital health in detection and management of atrial fibrillation. Heart, 108(11), 834–839.
<https://doi.org/10.1136/heartjnl-2020-318262>
- Torres-Soto, J., & Ashley, E. A. (2020). Multi-task deep learning for cardiac rhythm detection in wearable devices. Npj Digital Medicine, 3(1), 116.
<https://doi.org/10.1038/s41746-020-00320-4>
- Tran, H. H.-V., Urgessa, N. A., Geethakumari, P., Kampa, P., Parchuri, R., Bhandari, R., Alnasser, A. R., Akram, A., Kar, S., Osman, F., Mashat, G. D., & Mohammed, L. (2023). Detection and Diagnostic Accuracy of Cardiac Arrhythmias Using Wearable Health Devices: A Systematic Review. Cureus.
<https://doi.org/10.7759/cureus.50952>

- Tseng, A. S., & Noseworthy, P. A. (2021). Prediction of Atrial Fibrillation Using Machine Learning: A Review. *Frontiers in Physiology*, 12, 752317. <https://doi.org/10.3389/fphys.2021.752317>
- Turakhia, M. P., Desai, M., Hedlin, H., Rajmane, A., Talati, N., Ferris, T., Desai, S., Nag, D., Patel, M., Kowey, P., Rumsfeld, J. S., Russo, A. M., Hills, M. T., Granger, C. B., Mahaffey, K. W., & Perez, M. V. (2019). Rationale and design of a large-scale, app-based study to identify cardiac arrhythmias using a smartwatch: The Apple Heart Study. *American Heart Journal*, 207, 66–75. <https://doi.org/10.1016/j.ahj.2018.09.002>
- Vargova, E., Nemcova, A., Smisek, R., & Novakova, Z. (2024, December 1). Going beyond Atrial Fibrillation in Arrhythmia Classification from Photoplethysmography Signals. 2024 Computing in Cardiology Conference. <https://doi.org/10.22489/CinC.2024.110>
- Virani, S. S., Alonso, A., Aparicio, H. J., Benjamin, E. J., Bittencourt, M. S., Callaway, C. W., Carson, A. P., Chamberlain, A. M., Cheng, S., Delling, F. N., Elkind, M. S. V., Evenson, K. R., Ferguson, J. F., Gupta, D. K., Khan, S. S., Kissela, B. M., Knutson, K. L., Lee, C. D., Lewis, T. T., ... American Heart Association Council on Epidemiology and Prevention Statistics Committee and Stroke Statistics Subcommittee. (2021). Heart Disease and Stroke Statistics-2021 Update: A Report From the American Heart Association. *Circulation*, 143(8), e254–e743. <https://doi.org/10.1161/CIR.0000000000000950>

- Voisin, M., Shen, Y., Aliamiri, A., Avati, A., Hannun, A., & Ng, A. (2018). Ambulatory Atrial Fibrillation Monitoring Using Wearable Photoplethysmography with Deep Learning (arXiv:1811.07774). arXiv. <https://doi.org/10.48550/arXiv.1811.07774>
- Whiting, S., Moreland, S., Costello, J., Colopy, G., & McCann, C. (2018). Recognising Cardiac Abnormalities in Wearable Device Photoplethysmography (PPG) with Deep Learning (arXiv:1807.04077). arXiv. <https://doi.org/10.48550/arXiv.1807.04077>
- Wu, Y., Tang, Q., Zhan, W., Li, S., & Chen, Z. (2024). Res-BiANet: A Hybrid Deep Learning Model for Arrhythmia Detection Based on PPG Signal. *Electronics*, 13(3). <https://doi.org/10.3390/electronics13030665>
- Xie, H., Yang, L., Jiang, B., Huang, Z., & Lin, Y. (2025). State-of-the-art wearable sensors for cardiovascular health: A review. *Npj Cardiovascular Health*, 2, 53. <https://doi.org/10.1038/s44325-025-00090-6>
- Yan, R., Ding, C., Xiao, R., Fedorov, A., Lee, R. J., Nahab, F., & Hu, X. (2024). SQUWA: Signal Quality Aware DNN Architecture for Enhanced Accuracy in Atrial Fibrillation Detection from Noisy PPG Signals (arXiv:2404.15353). arXiv. <https://doi.org/10.48550/arXiv.2404.15353>
- Zhang, H., Zhu, L., Nathan, V., Kuang, J., Kim, J., Gao, J. A., & Olgin, J. (2021). Towards Early Detection and Burden Estimation of Atrial Fibrillation in an Ambulatory Free-living Environment. *Proceedings of the ACM on Interactive,*

Mobile, Wearable and Ubiquitous Technologies, 5(2), 1–19.
<https://doi.org/10.1145/3463503>

Zhang, Y., Xu, J., Xie, M., Zhu, D., Song, H., & Wang, W. (2023). Efficient and Direct Inference of Heart Rate Variability using Both Signal Processing and Machine Learning (Version 1). arXiv. <https://doi.org/10.48550/ARXIV.2303.13637>

Zheng, Y., Wu, C., Cai, P., Zhong, Z., Huang, H., & Jiang, Y. (2023). Tiny-PPG: A Lightweight Deep Neural Network for Real-Time Detection of Motion Artifacts in Photoplethysmogram Signals on Edge Devices (Version 3). arXiv. <https://doi.org/10.48550/ARXIV.2305.03308>

Zhou, T., Li, H., Shen, Z., & Jia, Z. (2022). End-Edge Coordinated Multiview Deep Learning for Time-Efficient Atrial Fibrillation Detection. 2022 IEEE 24th Int Conf on High Performance Computing & Communications; 8th Int Conf on Data Science & Systems; 20th Int Conf on Smart City; 8th Int Conf on Dependability in Sensor, Cloud & Big Data Systems & Application (HPCC/DSS/SmartCity/DependSys), 2003–2010.
<https://doi.org/10.1109/HPCC-DSS-SmartCity-DependSys57074.2022.00298>

Zhu, L., Nathan, V., Kuang, J., Kim, J., Avram, R., Olgin, J., & Gao, J. (2022). Atrial Fibrillation Detection and Atrial Fibrillation Burden Estimation via Wearables. IEEE Journal of Biomedical and Health Informatics, 26(5), 2063–2074.
<https://doi.org/10.1109/JBHI.2021.313198>